

2

The Vlasov, two-fluid, and MHD models of plasma dynamics

2.1 Overview

We begin this chapter by developing the concept of conservation of particles in phase-space and then use this concept as the basis for establishing the three main models of plasma dynamics, namely Vlasov theory, two-fluid theory, and magnetohydrodynamics (MHD). The Vlasov model is the most detailed and characterizes plasma dynamics by following the temporal evolution of electron and ion velocity distribution functions. The two-fluid model is intermediate in complexity and approximates plasma as a system of mutually interacting, finite-pressure electron and ion fluids. The MHD model is the least detailed and approximates plasma as a single, finite-pressure, electrically conducting fluid. The question of which of these models to use when analyzing a given situation is essentially a matter of selecting the best tool for the task and furthermore, just as a mechanic might alternate between using a screwdriver and a pair of pliers for a specific task, it is often advantageous to alternate between these models when analyzing a specific problem. As we develop these three models, we will also take the opportunity to explore some immediate and important fundamental consequences of these models, most notably the strong dependence of a collisionless plasma on its past history (Vlasov model) and the freezing of magnetic flux into the arbitrarily moving frame of a perfectly conducting plasma (MHD). The chapter concludes with an examination of some applications having a close association with the derivation of these models; these are classical transport of plasma across a magnetic field, sheaths at the edge of a plasma bounded by a conducting wall, and Langmuir probes, a simple method for diagnosing plasmas.

2.2 Phase-space

Consider a particle moving in a one-dimensional space and let the position of the particle be $x = x(t)$ and the velocity of the particle be $v = v(t)$. A way to visualize

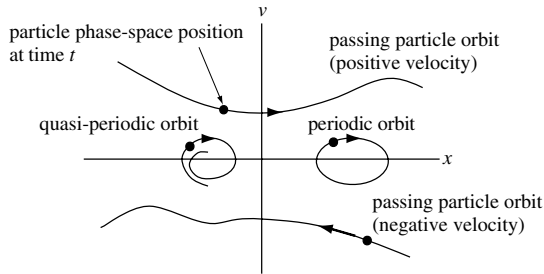


Fig. 2.1 Phase-space showing different types of possible particle orbits.

the x and v trajectories simultaneously is to plot these trajectories parametrically on a two-dimensional graph, where the horizontal coordinate is given by $x(t)$ and the vertical coordinate is given by $v(t)$. This $x - v$ plane is called *phase-space*. The trajectory (or orbit) of several particles can be represented as a set of curves in phase-space as shown in Fig. 2.1. Examples of a few qualitatively different phase-space orbits are shown in Fig. 2.1.

Particles in the upper half-plane always move to the right, since they have a positive velocity, while those in the lower half-plane always move to the left. Particles having exact periodic motion (e.g., $x = A \cos[\omega t]$, $v = -\omega A \sin[\omega t]$) alternate between moving to the right and the left and so describe an ellipse in phase-space. Particles with nearly periodic (quasi-periodic) motions will have near-ellipses or spiral orbits. A particle that does not reverse direction is called a passing particle, while a particle confined to a certain region of phase-space (e.g., a particle with periodic motion) is called a trapped particle.

2.3 Distribution function and Vlasov equation

At any given time, each particle has a specific position and velocity. We can therefore characterize the instantaneous configuration of a large number of particles by specifying the density of particles at each point x, v in phase-space. The function prescribing the instantaneous density of particles in phase-space is called the *distribution function* and is denoted by $f(x, v, t)$. Thus, $f(x, v, t)dx dv$ is the number of particles at time t having positions in the range between x and $x + dx$ and velocities in the range between v and $v + dv$. As time progresses, the particle motion and acceleration causes the number of particles in these x and v ranges to change and so f will change. This temporal evolution of f gives a description of the system more detailed than a fluid description, but less detailed than following the trajectory of each individual particle. Using the evolution of f to characterize

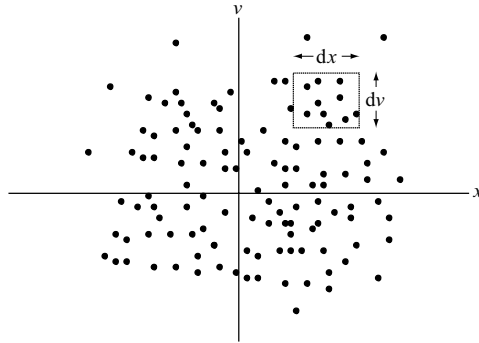


Fig. 2.2 A box within phase-space having width dx and height dv .

the system does not keep track of the trajectories of individual particles, but rather characterizes classes of particles having the same x, v .

Now consider the rate of change of the number of particles inside a small box in phase-space, such as is shown in Fig. 2.2. Defining $a(x, v, t)$ to be the acceleration of a particle, it is seen that the particle flux in the horizontal direction is fv and the particle flux in the vertical direction is fa . Thus, the particle fluxes into the four sides of the box are:

1. flux into left side of box is $f(x, v, t)vdv$,
2. flux into right side of box is $-f(x + dx, v, t)vdv$,
3. flux into bottom of box is $f(x, v, t)a(x, v, t)dx$,
4. flux into top of box is $-f(x, v + dv, t)a(x, v + dv, t)dx$.

The number of particles in the box is $f(x, v, t)dx dv$ so that the rate of change of the number of particles in the box is

$$\begin{aligned} \frac{\partial f(x, v, t)}{\partial t} dx dv = & -f(x + dx, v, t)vdv + f(x, v, t)vdv \\ & -f(x, v + dv, t)a(x, v + dv, t)dx \\ & +f(x, v, t)a(x, v, t)dx \end{aligned} \quad (2.1)$$

or, on Taylor expanding the quantities on the right-hand side, we obtain the one-dimensional *Vlasov equation*,

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} + \frac{\partial}{\partial v} (af) = 0. \quad (2.2)$$

It is straightforward to generalize Eq. (2.2) to three dimensions and so obtain the three-dimensional Vlasov equation,

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{a}f) = 0. \quad (2.3)$$

Because $\mathbf{x}$, $\mathbf{v}$ are independent quantities in phase-space, the spatial derivative term has the commutation property,

$$\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} = \frac{\partial}{\partial \mathbf{x}} \cdot (\mathbf{v} f). \quad (2.4)$$

The particle acceleration is given by the Lorentz force

$$\mathbf{a} = \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (2.5)$$

Because $(\mathbf{v} \times \mathbf{B})_i = v_j B_k - v_k B_j$ is independent of v_i , the term $\partial(\mathbf{v} \times \mathbf{B})_i / \partial v_i$ vanishes so that even though the Lorentz acceleration $\mathbf{a}$ is velocity-dependent, it nevertheless commutes with the vector velocity derivative as

$$\mathbf{a} \cdot \frac{\partial f}{\partial \mathbf{v}} = \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{a} f). \quad (2.6)$$

Because of this commutation property the Vlasov equation can also be written as

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{a} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (2.7)$$

If we “sit on top of” a particle that has a phase-space trajectory $\mathbf{x} = \mathbf{x}(t)$, $\mathbf{v} = \mathbf{v}(t)$ and measure the distribution function as we move along with the particle, the observed rate of change of the distribution function will be $df(\mathbf{x}(t), \mathbf{v}(t), t)/dt$, where the d/dt means that the derivative is measured in the moving frame. Because $d\mathbf{x}/dt = \mathbf{v}$ and $d\mathbf{v}/dt = \mathbf{a}$, this observed rate of change is

$$\left(\frac{df(\mathbf{x}(t), \mathbf{v}(t), t)}{dt} \right)_{\text{orbit}} = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{a} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \quad (2.8)$$

Thus, the distribution function f as measured when moving along a particle trajectory (orbit) is *constant*. This gives a powerful method for finding solutions to the Vlasov equation. Since f is a constant when measured in a frame following an orbit, we can choose f to depend on *any quantity* that is constant along the orbit (Jeans 1915, Watson 1956).

For example, if the energy E of particles is constant along their orbits then $f = f(E)$ is a solution to the Vlasov equation. On the other hand, if both the energy and the momentum p are constant along particle orbits, then any distribution function with the functional dependence $f = f(E, p)$ is a solution to the Vlasov equation. Depending on the situation at hand, the energy and/or momentum may or may not be constant along an orbit and so whether or not $f = f(E, p)$ is a solution to the Vlasov equation depends on the specific problem under consideration. However, there always exists at least one constant of the motion for any trajectory because, just like every human being has an invariant birthday, the initial conditions of a particle trajectory are invariant along this trajectory. As a simple example, consider a situation where there is no electromagnetic field

so that $\mathbf{a} = 0$, in which case the particle trajectories are simply $\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t$, $\mathbf{v}(t) = \mathbf{v}_0$, where $\mathbf{x}_0, \mathbf{v}_0$ are the initial position and velocity. Let us check to see whether $f(\mathbf{x}_0)$ is a solution to the Vlasov equation. By writing $\mathbf{x}_0 = \mathbf{x}(t) - \mathbf{v}_0 t$ so $f(\mathbf{x}_0) = f(\mathbf{x}(t) - \mathbf{v}_0 t)$ we observe that indeed $f = f(\mathbf{x}_0)$ is a solution, since

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{a} \cdot \frac{\partial f}{\partial \mathbf{v}} = -\mathbf{v}_0 \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} = 0. \quad (2.9)$$

2.4 Moments of the distribution function

Let us count the particles in the shaded vertical strip in Fig. 2.3. The number of particles in this strip is the number of particles lying between x and $x + dx$, where x is the location of the left-hand side of the strip and $x + dx$ is the location of the right-hand side. The number of particles in the strip is equivalently defined as $n(x, t)dx$, where $n(x)$ is the *density* of particles at x . Thus we see that $\int f(x, v)dv = n(x)$; the transition from a phase-space description (i.e., x, v are independent variables) to a conventional description (i.e., only x is an independent variable) involves “integrating out” the velocity dependence to obtain a quantity (e.g., density) depending only on position. Since the number of particles is finite, and since f is a positive quantity, f must vanish as $v \rightarrow \pm\infty$.

Another way of viewing f is to consider $f(x, v, t)/n(x, t)$ as the probability that a randomly selected particle at position x has the velocity v at time t . Using this point of view, we see that averaging over the velocities of all particles at x gives the mean velocity $u(x, t) = \int dv v f(x, v, t)/n(x, t)$. Similarly, multiplying $f(x, v, t)/n(x, t)$ by $mv^2/2$ and integrating over velocity will give an expression for the mean kinetic energy of all the particles. This procedure of multiplying f by various powers of v and then integrating over velocity is called *taking moments of the distribution function*.

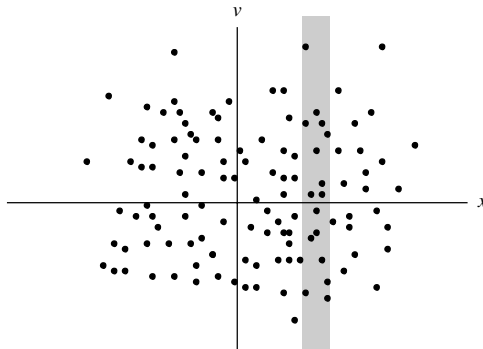


Fig. 2.3 Moments give weighted averages of the particles in the shaded vertical strip.

It is straightforward to generalize this “moment-taking” to three-dimensional problems simply by taking integrals over three-dimensional velocity space. Thus, in three dimensions the density becomes

$$n(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v} \quad (2.10)$$

and the mean velocity becomes

$$\mathbf{u}(\mathbf{x}, t) = \frac{\int \mathbf{v} f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v}}{n(\mathbf{x}, t)}. \quad (2.11)$$

2.4.1 Treatment of collisions in the Vlasov equation

It was shown in Section 1.8 that the cumulative effect of grazing collisions dominates the cumulative effect of the more infrequently occurring large-angle collisions. In order to see how collisions affect the Vlasov equation, let us now temporarily imagine that the grazing collisions are replaced by an equivalent sequence of abrupt large scattering angle encounters as shown in Fig. 2.4. Two particles involved in a collision do not significantly change their positions during the course of a collision, but they do substantially change their velocities. For example, a particle making a head-on collision with an equal mass stationary particle will stop after the collision, while the target particle will assume the velocity of the incident particle. If we draw the detailed phase-space trajectories characterized by a collision between two particles we see that each particle has a sudden change in its vertical coordinate (i.e., velocity) but no change in its horizontal coordinate (i.e., position). The collision-induced velocity jump occurs very fast so that if the phase-space trajectories were recorded with a “movie

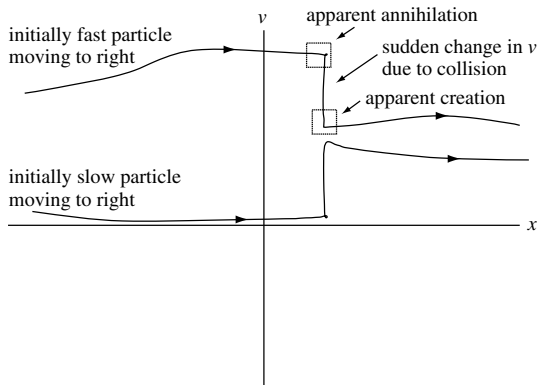


Fig. 2.4 Detailed view of collisions causing “jumps” in phase-space.

camera” having insufficient framing rate to catch the details of the jump, the resulting movie would show particles being spontaneously created or annihilated within given volumes of phase-space (e.g., within the boxes shown in Fig. 2.4).

The details of these individual jumps in phase-space are complicated and yet of little interest since all we really want to know is the cumulative effect of many collisions. It is therefore both efficient and sufficient to follow the trajectories on the slow time scale while accounting for the apparent “creation” or “annihilation” of particles by inserting a *collision operator* on the right-hand side of the Vlasov equation. In the example shown here it is seen that when a particle is apparently “created” in one box, another particle must be simultaneously “annihilated” in another box at the same x coordinate but a different v coordinate (of course, what is actually happening is that a single particle is suddenly moving from one box to the other). This coupling of the annihilation and creation rates in different boxes constrains the form of the collision operator. We will not attempt to derive collision operators in this chapter but will simply discuss the constraints on these operators. From a more formal point of view, collisions are characterized by constrained sources and sinks for particles in phase-space and inclusion of collisions in the Vlasov equation causes the Vlasov equation to assume the form

$$\frac{\partial f_\sigma}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (\mathbf{v} f_\sigma) + \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{a} f_\sigma) = \sum_\alpha C_{\sigma\alpha}(f_\sigma), \quad (2.12)$$

where $C_{\sigma\alpha}(f_\sigma)$ is the rate of change of f_σ due to collisions of species σ with species α .

The constraints that must be satisfied by the collision operator $C_{\sigma\alpha}(f_\sigma)$ are as follows:

- Conservation of particles – Collisions cannot change the total number of particles at a particular location so

$$\int d\mathbf{v} C_{\sigma\alpha}(f_\sigma) = 0. \quad (2.13)$$

- Conservation of momentum – Collisions between particles of the same species cannot change the total momentum of that species so

$$\int d\mathbf{v} m_\sigma \mathbf{v} C_{\sigma\sigma}(f_\sigma) = 0 \quad (2.14)$$

while collisions between different species must conserve the total momentum of both species together so

$$\int d\mathbf{v} m_i \mathbf{v} C_{ie}(f_i) + \int d\mathbf{v} m_e \mathbf{v} C_{ei}(f_e) = 0. \quad (2.15)$$

- Conservation of energy – Collisions between particles of the same species cannot change the total energy of that species so

$$\int d\mathbf{v} m_{\sigma} v^2 C_{\sigma\sigma}(f_{\sigma}) = 0 \quad (2.16)$$

while collisions between different species must conserve the total energy of both species together so

$$\int d\mathbf{v} m_i v^2 C_{ie}(f_i) + \int d\mathbf{v} m_e v^2 C_{ei}(f_e) = 0. \quad (2.17)$$

2.5 Two-fluid equations

Instead of just taking moments of the distribution function f itself, moments will now be taken of the entire Vlasov equation to obtain a set of partial differential equations relating the mean quantities $n(\mathbf{x})$, $\mathbf{u}(\mathbf{x})$, etc. We begin by integrating the Vlasov equation, Eq. (2.12), over velocity for each species. This first and simplest step in the procedure is called taking the “zeroth” moment, since the operation of multiplying by unity can be considered as multiplying the entire Vlasov equation by $\mathbf{v}$ raised to the power zero. Multiplying the Vlasov equation by unity and then integrating over velocity gives

$$\int \left[\frac{\partial f_{\sigma}}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (\mathbf{v} f_{\sigma}) + \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{a} f_{\sigma}) \right] d\mathbf{v} = \sum_{\alpha} \int C_{\sigma\alpha}(f_{\sigma}) d\mathbf{v}. \quad (2.18)$$

The velocity integral commutes with both the time and space derivatives on the left-hand side because $\mathbf{x}$, $\mathbf{v}$, and t are independent variables, while the third term on the left-hand side is the volume integral of a divergence in velocity space. Gauss’ theorem (i.e., $\int_{vol} d\mathbf{x} \nabla \cdot \mathbf{Q} = \int_{sfc} d\mathbf{s} \cdot \mathbf{Q}$) gives f_{σ} evaluated on a surface at $v = \infty$. However, because $f_{\sigma} \rightarrow 0$ as $v \rightarrow \infty$, this surface integral in velocity space vanishes. Using Eqs. (2.10), (2.11), and (2.13), we see that Eq. (2.18) becomes the species *continuity equation*

$$\frac{\partial n_{\sigma}}{\partial t} + \nabla \cdot (n_{\sigma} \mathbf{u}_{\sigma}) = 0. \quad (2.19)$$

Now let us multiply Eq. (2.12) by $\mathbf{v}$ and integrate over velocity to take the “first moment,”

$$\int \mathbf{v} \left[\frac{\partial f_{\sigma}}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (\mathbf{v} f_{\sigma}) + \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{a} f_{\sigma}) \right] d\mathbf{v} = \sum_{\alpha} \int \mathbf{v} C_{\sigma\alpha}(f_{\sigma}) d\mathbf{v}. \quad (2.20)$$

This may be rearranged in a more tractable form by:

- (i) “pulling” both the time and space derivatives out of the velocity integral,
- (ii) writing $\mathbf{v} = \mathbf{v}'(\mathbf{x}, t) + \mathbf{u}(\mathbf{x}, t)$, where $\mathbf{v}'(\mathbf{x}, t)$ is the *random* part of a given velocity, i.e., that part of the velocity that differs from the mean (note that $\mathbf{v}$ is independent of both $\mathbf{x}$ and t but $\mathbf{v}'$ is not; also $d\mathbf{v} = d\mathbf{v}'$),
- (iii) integrating by parts in 3-D velocity space on the acceleration term and using

$$\left(\frac{\partial \mathbf{v}}{\partial \mathbf{v}} \right)_{ij} = \delta_{ij}.$$

After performing these manipulations, the first moment of the Vlasov equation becomes

$$\begin{aligned} \frac{\partial (n_\sigma \mathbf{u}_\sigma)}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot \int (\mathbf{v}' \mathbf{v}' + \mathbf{v}' \mathbf{u}_\sigma + \mathbf{u}_\sigma \mathbf{v}' + \mathbf{u}_\sigma \mathbf{u}_\sigma) f_\sigma d\mathbf{v}' \\ - \frac{q_\sigma}{m_\sigma} \int (\mathbf{E} + \mathbf{v} \times \mathbf{B}) f_\sigma d\mathbf{v}' = -\frac{1}{m_\sigma} \mathbf{R}_{\sigma\alpha}, \end{aligned} \quad (2.21)$$

where $\mathbf{R}_{\sigma\alpha}$ is the net frictional drag force due to collisions of species σ with species α . Note that $\mathbf{R}_{\sigma\sigma} = 0$, since a species cannot exert a net drag force on itself. This is because, just like one cannot pull oneself up by one’s own bootstraps, the totality of electrons cannot cause a force that changes the center-of-mass velocity of the totality of electrons. The frictional terms have the form

$$\mathbf{R}_{ei} = \nu_{ei} m_e n_e (\mathbf{u}_e - \mathbf{u}_i), \quad (2.22)$$

$$\mathbf{R}_{ie} = \nu_{ie} m_i n_i (\mathbf{u}_i - \mathbf{u}_e), \quad (2.23)$$

so that in the ion frame the drag on electrons is simply the total electron momentum $m_e n_e \mathbf{u}_e$ measured in this frame multiplied by the rate ν_{ei} at which this momentum is destroyed by collisions with ions. This form for frictional drag force has the following properties: (i) $\mathbf{R}_{ei} + \mathbf{R}_{ie} = 0$, showing that the plasma cannot exert a frictional drag force on itself, (ii) friction causes the faster species to be slowed down by the slower species, and (iii) there is no friction between species if both have the same mean velocity.

Equation (2.21) can be further simplified by factoring $\mathbf{u}$ out of the velocity integrals and recalling that by definition $\int \mathbf{v}' f_\sigma d\mathbf{v}' = 0$. Thus, Eq. (2.21) reduces to

$$m_\sigma \left[\frac{\partial (n_\sigma \mathbf{u}_\sigma)}{\partial t} + \frac{\partial}{\partial \mathbf{x}} \cdot (n_\sigma \mathbf{u}_\sigma \mathbf{u}_\sigma) \right] = n_\sigma q_\sigma (\mathbf{E} + \mathbf{u}_\sigma \times \mathbf{B}) - \frac{\partial}{\partial \mathbf{x}} \cdot \overleftrightarrow{\mathbf{P}}_\sigma - \mathbf{R}_{\sigma\alpha}, \quad (2.24)$$

where the *pressure tensor* $\overleftrightarrow{\mathbf{P}}$ is defined by

$$\overleftrightarrow{\mathbf{P}}_\sigma = m_\sigma \int \mathbf{v}' \mathbf{v}' f_\sigma d\mathbf{v}'.$$

If f_σ is an isotropic function of $\mathbf{v}'$, then the off-diagonal terms in $\overleftrightarrow{\mathbf{P}}_\sigma$ vanish and the three diagonal terms are identical. In this case, it is useful to define the diagonal terms to be the *scalar pressure* P_σ , i.e.,

$$\begin{aligned} P_\sigma &= m_\sigma \int v'_x v'_x f_\sigma d\mathbf{v}' = m_\sigma \int v'_y v'_y f_\sigma d\mathbf{v}' = m_\sigma \int v'_z v'_z f_\sigma d\mathbf{v}' \\ &= \frac{m_\sigma}{3} \int \mathbf{v}' \cdot \mathbf{v}' f_\sigma d\mathbf{v}'. \end{aligned} \quad (2.25)$$

Equation (2.25) defines pressure for a three-dimensional isotropic system. However, we will often deal with systems of reduced dimensionality, i.e., systems with just one or two dimensions. Equation (2.25) can therefore be generalized to these other cases by introducing the general N -dimensional definition for scalar pressure

$$P_\sigma = \frac{m_\sigma}{N} \int \mathbf{v}' \cdot \mathbf{v}' f_\sigma d^N \mathbf{v}' = \frac{m_\sigma}{N} \int \sum_{j=1}^N v_j'^2 f_\sigma d^N \mathbf{v}', \quad (2.26)$$

where $\mathbf{v}'$ is the N -dimensional random velocity.

It is important to emphasize that assuming isotropy is done largely for mathematical convenience and that in real systems the distribution function is often quite anisotropic. Collisions, being randomizing, drive the distribution function towards isotropy, while competing processes simultaneously drive it towards anisotropy. Thus, each situation must be considered individually in order to determine whether there is sufficient collisionality to make f isotropic. Because fully ionized hot plasmas often have insufficient collisions to make f isotropic, the oft-used assumption of isotropy is an oversimplification, which may or may not be acceptable depending on the phenomenon under consideration.

On expanding the derivatives on the left-hand side of Eq. (2.24), it is seen that two of the terms combine to give $\mathbf{u}$ times Eq. (2.19). After removing this embedded continuity equation, Eq. (2.24) reduces to

$$n_\sigma m_\sigma \frac{d\mathbf{u}_\sigma}{dt} = n_\sigma q_\sigma (\mathbf{E} + \mathbf{u}_\sigma \times \mathbf{B}) - \nabla P_\sigma - \mathbf{R}_{\sigma\alpha}, \quad (2.27)$$

where the operator d/dt is defined to be the *convective derivative*

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{u}_\sigma \cdot \nabla, \quad (2.28)$$

which characterizes the temporal rate of change seen by an observer *moving with the mean fluid velocity* $\mathbf{u}_\sigma$ of species σ . An everyday example of the convective term would be the apparent temporal increase in density of automobiles seen by a motorcyclist who enters a traffic jam of stationary vehicles and is not impeded by the traffic jam.

At this point in the procedure it becomes evident that a certain pattern recurs for each successive moment of the Vlasov equation. When we took the zeroth moment, an equation for the density $\int f_\sigma d\mathbf{v}$ resulted, but this also introduced a term involving the next higher moment, namely the mean velocity $\sim \int \mathbf{v} f_\sigma d\mathbf{v}$. Then, when we took the first moment to get an equation for the velocity, an equation was obtained containing a term involving the next higher moment, namely the pressure $\sim \int \mathbf{v} \mathbf{v} f_\sigma d\mathbf{v}$. Thus, each time we take a moment of the Vlasov equation, an equation for the moment we want is obtained, but because of the $\mathbf{v} \cdot \nabla f$ term in the Vlasov equation, a next higher moment also appears. Thus, moment-taking never leads to a closed system of equations; there will always be a “loose end,” a highest moment for which there is no determining equation. Some sort of ad hoc closure procedure must always be invoked to terminate this chain (as seen below, typical closures involve invoking adiabatic or isothermal assumptions). Another feature of taking moments is that each higher moment embeds terms that contain complete lower moment equations multiplied by some factor. Algebraic manipulation can identify these lower moment equations and eliminate them to give a simplified higher moment equation.

Let us now take the second moment of the Vlasov equation. Unlike the zeroth and first moments, the dimensionality of the system now enters explicitly so the more general pressure definition given by Eq. (2.26) will be used. Multiplying the Vlasov equation by $m_\sigma v^2/2$ and integrating over velocity gives

$$\left\{ \begin{aligned} & \frac{\partial}{\partial t} \int \frac{m_\sigma v^2}{2} f_\sigma d^N \mathbf{v} \\ & + \frac{\partial}{\partial \mathbf{x}} \cdot \int \frac{m_\sigma v^2}{2} \mathbf{v} f_\sigma d^N \mathbf{v} \\ & + q_\sigma \int \frac{v^2}{2} \frac{\partial}{\partial \mathbf{v}} \cdot (\mathbf{E} + \mathbf{v} \times \mathbf{B}) f_\sigma d^N \mathbf{v} \end{aligned} \right\} = \sum_\alpha \int m_\sigma \frac{v^2}{2} C_{\sigma\alpha} f_\sigma d^N \mathbf{v}. \quad (2.29)$$

We consider each term of this equation separately as follows:

1. The time derivative term becomes

$$\frac{\partial}{\partial t} \int \frac{m_\sigma v^2}{2} f_\sigma d^N \mathbf{v} = \frac{\partial}{\partial t} \int \frac{m_\sigma (\mathbf{v}' + \mathbf{u}_\sigma)^2}{2} f_\sigma d^N \mathbf{v}' = \frac{\partial}{\partial t} \left(\frac{NP_\sigma}{2} + \frac{m_\sigma n_\sigma u_\sigma^2}{2} \right).$$

2. Again using $\mathbf{v} = \mathbf{v}' + \mathbf{u}_\sigma$ the space derivative term becomes

$$\frac{\partial}{\partial \mathbf{x}} \cdot \int \frac{m_\sigma v^2}{2} \mathbf{v} f_\sigma d^N \mathbf{v} = \nabla \cdot \left(\mathbf{Q}_\sigma + \frac{2+N}{2} P_\sigma \mathbf{u}_\sigma + \frac{m_\sigma n_\sigma u_\sigma^2}{2} \mathbf{u}_\sigma \right),$$

where $\mathbf{Q}_\sigma = \int \frac{m_\sigma v'^2}{2} \mathbf{v}' f_\sigma d^N \mathbf{v}$ is called the *heat flux*.

3. On integrating by parts, the acceleration term becomes

$$q_\sigma \int \frac{v^2}{2} \frac{\partial}{\partial \mathbf{v}} \cdot [(\mathbf{E} + \mathbf{v} \times \mathbf{B}) f_\sigma] d^N \mathbf{v} = -q_\sigma \int \mathbf{v} \cdot \mathbf{E} f_\sigma d\mathbf{v} = -q_\sigma n_\sigma \mathbf{u}_\sigma \cdot \mathbf{E}.$$

4. The collision term becomes (using Eq. (2.16))

$$\sum_\alpha \int m_\sigma \frac{v^2}{2} C_{\sigma\alpha} f_\sigma d\mathbf{v} = \int_{\alpha \neq \sigma} m_\sigma \frac{v^2}{2} C_{\sigma\alpha} f_\sigma d\mathbf{v} = - \left(\frac{\partial W}{\partial t} \right)_{E\sigma\alpha},$$

where $(\partial W/\partial t)_{E\sigma\alpha}$ is the rate at which species σ collisionally transfers energy to species α .

Combining the above four relations, Eq. (2.29) becomes

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{NP_\sigma}{2} + \frac{m_\sigma n_\sigma u_\sigma^2}{2} \right) + \nabla \cdot \left(\mathbf{Q}_\sigma + \frac{2+N}{2} P_\sigma \mathbf{u}_\sigma + \frac{m_\sigma n_\sigma u_\sigma^2}{2} \mathbf{u}_\sigma \right) - q_\sigma n_\sigma \mathbf{u}_\sigma \cdot \mathbf{E} \\ = - \left(\frac{\partial W}{\partial t} \right)_{E\sigma\alpha}. \end{aligned} \quad (2.30)$$

This equation can be simplified by invoking two mathematical identities, the first being

$$\frac{\partial}{\partial t} \left(\frac{m_\sigma n_\sigma u_\sigma^2}{2} \right) + \nabla \cdot \left(\frac{m_\sigma n_\sigma u_\sigma^2}{2} \mathbf{u}_\sigma \right) = n_\sigma \left(\frac{\partial}{\partial t} + \mathbf{u}_\sigma \cdot \nabla \right) \frac{m_\sigma u_\sigma^2}{2} = n_\sigma \frac{d}{dt} \left(\frac{m_\sigma u_\sigma^2}{2} \right). \quad (2.31)$$

The second identity is obtained by dotting the equation of motion with $\mathbf{u}_\sigma$ and is

$$\begin{aligned} n_\sigma m_\sigma \left[\frac{\partial}{\partial t} \left(\frac{u_\sigma^2}{2} \right) + \mathbf{u}_\sigma \cdot \left(\nabla \left(\frac{u_\sigma^2}{2} \right) - \mathbf{u}_\sigma \times \nabla \times \mathbf{u}_\sigma \right) \right] \\ = n_\sigma q_\sigma \mathbf{u}_\sigma \cdot \mathbf{E} - \mathbf{u}_\sigma \cdot \nabla P_\sigma - \mathbf{R}_{\sigma\alpha} \cdot \mathbf{u}_\sigma \end{aligned}$$

or

$$n_\sigma \frac{d}{dt} \left(\frac{m_\sigma u_\sigma^2}{2} \right) = n_\sigma q_\sigma \mathbf{u}_\sigma \cdot \mathbf{E} - \mathbf{u}_\sigma \cdot \nabla P_\sigma - \mathbf{R}_{\sigma\alpha} \cdot \mathbf{u}_\sigma. \quad (2.32)$$

Inserting Eqs. (2.31) and (2.32) in Eq. (2.30) gives the energy evolution equation

$$\frac{N}{2} \frac{dP_\sigma}{dt} + \frac{2+N}{2} P_\sigma \nabla \cdot \mathbf{u}_\sigma = -\nabla \cdot \mathbf{Q}_\sigma + \mathbf{R}_{\sigma\alpha} \cdot \mathbf{u}_\sigma - \left(\frac{\partial W}{\partial t} \right)_{E\sigma\alpha}. \quad (2.33)$$

The first term on the right-hand side represents the heat flux, the second term gives the frictional heating of species σ due to frictional drag on species α , while the last term on the right-hand side gives the rate at which species σ collisionally transfers energy to other species. Although Eq. (2.33) is complicated, two important limiting situations become evident if we let t_{char} be the characteristic time scale for a given phenomenon and l_{char} be its characteristic length scale. A characteristic velocity

$V_{ph} \sim l_{char}/t_{char}$ may then be defined for the phenomenon and so, replacing temporal derivatives by t_{char}^{-1} and spatial derivatives by l_{char}^{-1} in Eq. (2.33), it is seen that the two limiting situations are:

1. *Isothermal limit* – The heat flux term dominates all other terms, in which case the temperature becomes spatially uniform. This occurs if (i) $v_{T\sigma} \gg V_{ph}$ since the ratio of the left-hand side terms to the heat flux term is $\sim V_{ph}/v_{T\sigma}$ and (ii) the collisional terms are small enough to be ignored.
2. *Adiabatic limit* – The heat flux terms and the collisional terms are small enough to be ignored compared to the left-hand side terms; this occurs when $V_{ph} \gg v_{T\sigma}$. Adiabatic is a Greek word meaning “impassable,” and is used here to denote that no heat is flowing, i.e., the volume under consideration is thermally isolated from the outside world.

Both of these limits make it possible to avoid solving for $\mathbf{Q}_\sigma$, which involves the third moment, and so both the adiabatic and isothermal limit provide a closure to the moment equations.

The energy equation may be greatly simplified in the adiabatic limit by rearranging the continuity equation to give

$$\nabla \cdot \mathbf{u}_\sigma = -\frac{1}{n_\sigma} \frac{dn_\sigma}{dt} \quad (2.34)$$

and then substituting this expression into the left-hand side of Eq. (2.33) to obtain

$$\frac{1}{P_\sigma} \frac{dP_\sigma}{dt} = \frac{\gamma}{n_\sigma} \frac{dn_\sigma}{dt}, \quad (2.35)$$

where

$$\gamma = \frac{N+2}{N}. \quad (2.36)$$

Equation (2.35) implies

$$\frac{d}{dt} \left(\frac{P_\sigma}{n_\sigma^\gamma} \right) = 0 \quad (2.37)$$

so

$$\frac{P_\sigma}{n_\sigma^\gamma} = \text{constant in the frame of the moving plasma.} \quad (2.38)$$

This constitutes a derivation of adiabaticity based on geometry and statistical mechanics rather than on thermodynamic arguments.

2.5.1 Entropy of a distribution function

Collisions cause the distribution function to tend towards a simple final state characterized by having the maximum entropy for the given constraints (e.g., fixed

total energy). To see this, we provide a brief discussion of entropy and show how it relates to a distribution function.

Suppose we throw two dice, labeled A and B , and let R denote the result of a throw. Thus R ranges from 2 through 12. The complete set of (A, B) combinations that gives these R 's is listed below:

$$\begin{aligned} R = 2 &\iff (1,1) \\ R = 3 &\iff (1,2), (2,1) \\ R = 4 &\iff (1,3), (3,1), (2,2) \\ R = 5 &\iff (1,4), (4,1), (2,3), (3,2) \\ R = 6 &\iff (1,5), (5,1), (2,4), (4,2), (3,3) \\ R = 7 &\iff (1,6), (6,1), (2,5), (5,2), (3,4), (4,3) \\ R = 8 &\iff (2,6), (6,2), (3,5), (5,3), (4,4) \\ R = 9 &\iff (3,6), (6,3), (4,5), (5,4) \\ R = 10 &\iff (4,6), (6,4), (5,5) \\ R = 11 &\iff (5,6), (6,5) \\ R = 12 &\iff (6,6) \end{aligned}$$

There are six (A, B) pairs that give $R = 7$, but only one pair for $R = 2$ and only one pair for $R = 12$. Thus, there are six microscopic states (distinct (A, B) pairs) corresponding to $R = 7$ but only one microscopic state corresponding to each of $R = 2$ or $R = 12$. Thus, we *know more* about the microscopic state of the system if $R = 2$ or 12 than if $R = 7$. We define the entropy S to be the *natural logarithm of the number of microscopic states corresponding to a given macroscopic state*. Thus, for the dice, the entropy would be the natural logarithm of the number of (A, B) pairs that correspond to a given R . The entropy for $R = 2$ or $R = 12$ would be zero since $S = \ln(1) = 0$, while the entropy for $R = 7$ would be $S = \ln(6)$ since there were six different ways of obtaining $R = 7$.

If the dice were to be thrown a statistically large number of times the most likely result for any throw is $R = 7$; this is the macroscopic state with the largest number of microscopic states. Since any of the possible microscopic states is an equally likely outcome, the most likely macroscopic state after a large number of dice throws is the macroscopic state with the highest entropy.

Now consider a situation more closely related to the concept of a distribution function. In order to do this we first pose the following simple problem: suppose we have a pegboard with $\mathcal{N}$ holes, labeled $h_1, h_2, \dots, h_{\mathcal{N}}$, and we also have $\mathcal{N}$ pegs labeled by $p_1, p_2, \dots, p_{\mathcal{N}}$. What is the number of ways of putting all the pegs in all the holes? Starting with hole h_1 , we have a choice of $\mathcal{N}$ different pegs, but when we get to hole h_2 there are now only $\mathcal{N} - 1$ pegs remaining so that there are now only $\mathcal{N} - 1$ choices. Using this argument for subsequent holes, we see there are $\mathcal{N}!$ ways of putting all the pegs in all the holes.

Let us complicate things further. Suppose that we arrange the holes in $\mathcal{M}$ groups. Say group G_1 has the first 10 holes, group G_2 has the next 19 holes, group G_3 has the next 4 holes and so on, up to group $\mathcal{M}$. We will use f to denote the number of holes in a group, thus $f(1) = 10$, $f(2) = 19$, $f(3) = 4$, etc. The number of ways of arranging pegs within a group is just the factorial of the number of pegs in the group, e.g., the number of ways of arranging the pegs within group 1 is just $10!$ and so in general the number of ways of arranging the pegs in the j th group is $[f(j)]!$.

Let us denote C as the number of ways of putting all the pegs in all the groups *without caring* about the internal arrangement within groups. The number of ways of putting the pegs in all the groups *caring* about the internal arrangements in all the groups is $C \times f(1)! \times f(2)! \times \dots \times f(\mathcal{M})!$, but this is just the number of ways of putting all the pegs in all the holes, i.e.,

$$C \times f(1)! \times f(2)! \times \dots \times f(\mathcal{M})! = \mathcal{N}!$$

or

$$C = \frac{\mathcal{N}!}{f(1)! \times f(2)! \times \dots \times f(\mathcal{M})!}.$$

Now C is just the number of microscopic states corresponding to the macroscopic state of the prescribed grouping $f(1) = 10$, $f(2) = 19$, $f(3) = 4$, etc. so the entropy is just $S = \ln C$ or

$$\begin{aligned} S &= \ln \left(\frac{\mathcal{N}!}{f(1)! \times f(2)! \times \dots \times f(\mathcal{M})!} \right) \\ &= \ln \mathcal{N}! - \ln f(1)! - \ln f(2)! - \dots - \ln f(\mathcal{M})!. \end{aligned} \quad (2.39)$$

Stirling's formula shows that the large-argument asymptotic limit of the factorial function is

$$\lim_{k \rightarrow \text{large}} \ln k! = k \ln k - k. \quad (2.40)$$

Noting that $f(1) + f(2) + \dots + f(\mathcal{M}) = \mathcal{N}$ the entropy becomes

$$\begin{aligned} S &= \mathcal{N} \ln \mathcal{N} - f(1) \ln f(1) - f(2) \ln f(2) - \dots - f(\mathcal{M}) \ln f(\mathcal{M}) \\ &= \mathcal{N} \ln \mathcal{N} - \sum_{j=1}^{\mathcal{M}} f(j) \ln f(j). \end{aligned} \quad (2.41)$$

The constant $\mathcal{N} \ln \mathcal{N}$ is often dropped, giving

$$S = - \sum_{j=1}^{\mathcal{M}} f(j) \ln f(j). \quad (2.42)$$

If j is made into a continuous variable, say $j \rightarrow v$ so that $f(v)dv$ is the number of items in the group labeled by v , then the entropy can be written as

$$S = - \int dv f(v) \ln f(v). \quad (2.43)$$

By now, it is obvious that f could be the velocity distribution function, in which case $f(v)dv$ is just the number of particles in the group having velocity between v and $v + dv$. Since the peg groups correspond to different velocity range coordinates, having more dimensions just means having more groups and so for three dimensions the entropy generalizes to

$$S = - \int d\mathbf{v} f(\mathbf{v}) \ln f(\mathbf{v}). \quad (2.44)$$

If the distribution function depends on position as well, this corresponds to still more peg groups, and so a distribution function that depends on both velocity and position will have the entropy

$$S = - \int d\mathbf{x} \int d\mathbf{v} f(\mathbf{x}, \mathbf{v}) \ln f(\mathbf{x}, \mathbf{v}). \quad (2.45)$$

2.5.2 Effect of collisions on entropy

The highest entropy state is the most likely state of the system because the highest entropy state has the highest number of microscopic states corresponding to the macroscopic state. Collisions (or other forms of randomization) will take some initial prescribed microscopic state and scramble the phase-space positions of the particles, thereby transforming the system to a different microscopic state. This new state could in principle be any microscopic state, but is most likely to be a member of the class of microscopic states belonging to the highest entropy macroscopic state. Thus, any randomization process such as collisions will cause the system to evolve towards the macroscopic state having the maximum entropy.

An important shortcoming of this argument is that it neglects any conservation relations that have to be satisfied. To see this, note that the expression for entropy could be maximized if all the particles are put in one group, in which case $C = \mathcal{N}!$, which is the largest possible value for C . Thus, the maximum entropy configuration of $\mathcal{N}$ plasma particles corresponds to all the particles having the same velocity. However, this would assign a specific energy to the system, which would in general differ from the energy of the initial microstate. This maximum entropy state is therefore not accessible in isolated systems, because energy would not be conserved if the system changed from its initial microstate to the maximum entropy state.

Thus, a qualification must be added to the argument. Randomizing processes will scramble the system to attain the state of maximum entropy *consistent* with

any constraints placed on the system. Examples of such constraints would be the requirements that the total system energy and the total number of particles must be conserved. We therefore reformulate the problem as: given an isolated system with $\mathcal{N}$ particles in a fixed volume V and initial average energy per particle $\langle E \rangle$, what is the maximum entropy state consistent with conservation of energy and conservation of number of particles? This is a variational problem because the goal is to maximize S subject to the constraint that both $\mathcal{N}$ and $\mathcal{N} \langle E \rangle$ are fixed. The method of Lagrange multipliers can then be used to take into account these constraints (see Assignment 2 for a derivation of the Lagrange multiplier method). Using this method the variational problem becomes

$$\delta S - \lambda_1 \delta \mathcal{N} - \lambda_2 \delta (\mathcal{N} \langle E \rangle) = 0, \quad (2.46)$$

where λ_1 and λ_2 are as-yet undetermined Lagrange multipliers. The number of particles is

$$\mathcal{N} = V \int f dv. \quad (2.47)$$

The energy of an individual particle is $E = mv^2/2$, where v is the velocity measured in the rest frame of the center of mass of the entire collection of $\mathcal{N}$ particles. Thus, the total kinetic energy of all the particles in this rest frame is

$$\mathcal{N} \langle E \rangle = V \int \frac{mv^2}{2} f(v) dv \quad (2.48)$$

and so the variational problem becomes

$$\delta \int dv \left(f \ln f - \lambda_1 V f - \lambda_2 V \frac{mv^2}{2} f \right) = 0. \quad (2.49)$$

Incorporating the volume V into the Lagrange multipliers, and factoring out the coefficient δf , this becomes

$$\int dv \delta f \left(1 + \ln f - \lambda_1 - \lambda_2 \frac{mv^2}{2} \right) = 0. \quad (2.50)$$

Since δf is arbitrary, the integrand must vanish, giving

$$\ln f = \lambda_2 \frac{mv^2}{2} - \lambda_1, \quad (2.51)$$

where the “1” has been incorporated into λ_1 .

The maximum entropy distribution function of an isolated, energy and particle conserving system is therefore

$$f = \lambda_1 \exp(-\lambda_2 mv^2/2); \quad (2.52)$$

this is called a Maxwellian distribution function. We will often assume that the plasma is locally Maxwellian so that $\lambda_1 = \lambda_1(\mathbf{x}, t)$ and $\lambda_2 = \lambda(\mathbf{x}, t)$. We define the temperature to be

$$\kappa T_\sigma(\mathbf{x}, t) = \frac{1}{\lambda_2(\mathbf{x}, t)}, \quad (2.53)$$

where Boltzmann's factor κ allows temperature to be measured in various units. The normalization factor is set to be

$$\lambda_1(\mathbf{x}, t) = n(\mathbf{x}, t) \left(\frac{m_\sigma}{2\pi\kappa T_\sigma(\mathbf{x}, t)} \right)^{N/2}, \quad (2.54)$$

where N is the dimensionality (1, 2, or 3) so that $\int f_\sigma(\mathbf{x}, \mathbf{v}, t) d^N \mathbf{v} = n_\sigma(\mathbf{x}, t)$. Because the kinetic energy of individual particles was defined in terms of velocities measured in the rest frame of the center of mass of the complete system of particles, if this center of mass is moving in the lab frame with a velocity $\mathbf{u}_\sigma$, then in the lab frame the Maxwellian will have the form

$$f_\sigma(\mathbf{x}, \mathbf{v}, t) = n_\sigma \left(\frac{m_\sigma}{2\pi\kappa T_\sigma} \right)^{N/2} \exp(-m_\sigma(\mathbf{v} - \mathbf{u}_\sigma)^2 / 2\kappa T_\sigma). \quad (2.55)$$

2.5.3 Relation between pressure and Maxwellian

The scalar pressure has a simple relation to the generalized Maxwellian as seen by recasting Eq. (2.26) as

$$\begin{aligned} P_\sigma &= \frac{m_\sigma}{N} \int \mathbf{v}' \cdot \mathbf{v}' f_\sigma d^N \mathbf{v}' \\ &= \frac{n_\sigma m_\sigma}{N} \left(\frac{m_\sigma}{2\pi\kappa T_\sigma} \right)^{N/2} \int (\mathbf{v}')^2 \exp(-m_\sigma(\mathbf{v}')^2 / 2\kappa T_\sigma) d^N \mathbf{v}' \\ &= -\frac{n_\sigma m_\sigma}{N} \left(\frac{\alpha}{\pi} \right)^{N/2} \frac{d}{d\alpha} \int e^{-\alpha v'^2} d^N \mathbf{v}', \text{ defining } \alpha = m_\sigma / 2\kappa T_\sigma \\ &= -\frac{n_\sigma m_\sigma}{N} \left(\frac{\alpha}{\pi} \right)^{N/2} \frac{d}{d\alpha} \left(\frac{\alpha}{\pi} \right)^{-N/2} \\ &= n_\sigma \kappa T_\sigma, \end{aligned} \quad (2.56)$$

which is just the ideal gas law. Thus, the definitions that have been proposed for pressure and temperature are consistent with everyday notions for these quantities.

Despite this correspondence to familiar concepts, we must be careful not to become overconfident regarding the descriptive power of the fluid point of view because weaknesses exist in this point of view. For example, as discussed on p. 46 neither the adiabatic nor the isothermal approximation is appropriate when $V_{ph} \sim v_{T\sigma}$. The fluid description breaks down in this situation and, as will be

seen in a later chapter, the Vlasov description must be used in this situation. Furthermore, the distribution function is Maxwellian only if there are sufficient collisions or some other randomizing process. Because of the weak collisionality of a plasma, this is very often not the case. In particular, since the collision frequency scales as v^{-3} , fast particles take much longer to become Maxwellian than slow particles. It is not at all unusual for a real plasma to be in a state where the low-velocity particles have reached a Maxwellian distribution whereas the fast particles form a non-Maxwellian “tail.”

We now summarize the two-fluid equations:

- continuity equation for each species

$$\frac{\partial n_\sigma}{\partial t} + \nabla \cdot (n_\sigma \mathbf{u}_\sigma) = 0; \tag{2.57}$$

- equation of motion for each species

$$n_\sigma m_\sigma \frac{d\mathbf{u}_\sigma}{dt} = n_\sigma q_\sigma (\mathbf{E} + \mathbf{u}_\sigma \times \mathbf{B}) - \nabla P_\sigma - \mathbf{R}_{\sigma\alpha}; \tag{2.58}$$

- equation of state for each species

Regime	Equation of state	Name
$V_{ph} \gg v_{T\sigma}$	$P_\sigma \sim n_\sigma^\gamma$	adiabatic
$V_{ph} \ll v_{T\sigma}$	$P_\sigma = n_\sigma \kappa T_\sigma, T_\sigma = \text{constant}$	isothermal

- Maxwell’s equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.59}$$

$$\nabla \times \mathbf{B} = \mu_0 \sum_\sigma n_\sigma q_\sigma \mathbf{u}_\sigma + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.60}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2.61}$$

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \sum_\sigma n_\sigma q_\sigma. \tag{2.62}$$

2.6 Magnetohydrodynamic equations

Particle motion in the two-fluid system was described by the individual species’ mean velocities $\mathbf{u}_e, \mathbf{u}_i$ and by the pressures $\overleftrightarrow{\mathbf{P}}_e, \overleftrightarrow{\mathbf{P}}_i$, which provide an accounting for the mean square of the random deviation of the velocity from its average value. Magnetohydrodynamics is an alternate description of the plasma where,

instead of using $\mathbf{u}_e, \mathbf{u}_i$ to describe mean motion, two new velocity variables that are a linear combination of $\mathbf{u}_e, \mathbf{u}_i$ are used. As will be seen below, this means a slightly different definition for pressure must also be used.

The new velocity-like variables are (i) the current density

$$\mathbf{J} = \sum_{\sigma} n_{\sigma} q_{\sigma} \mathbf{u}_{\sigma}, \quad (2.63)$$

which is essentially the relative velocity between ions and electrons, and (ii) the center-of-mass velocity

$$\mathbf{U} = \frac{1}{\rho} \sum_{\sigma} m_{\sigma} n_{\sigma} \mathbf{u}_{\sigma}, \quad (2.64)$$

where

$$\rho = \sum_{\sigma} m_{\sigma} n_{\sigma} \quad (2.65)$$

is the total mass density. Magnetohydrodynamics is primarily concerned with low-frequency, long-wavelength, *magnetic* behavior of the plasma.

2.6.1 MHD continuity equation

Multiplying Eq. (2.19) by m_{σ} and summing over species gives the MHD continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0. \quad (2.66)$$

2.6.2 MHD equation of motion

To obtain an equation of motion we take the first moment of the Vlasov equation, then multiply by m_{σ} and sum over species to obtain

$$\frac{\partial}{\partial t} \sum_{\sigma} m_{\sigma} \int \mathbf{v} f_{\sigma} d\mathbf{v} + \frac{\partial}{\partial \mathbf{x}} \cdot \sum_{\sigma} \int m_{\sigma} \mathbf{v} \mathbf{v} f_{\sigma} d\mathbf{v} + \sum_{\sigma} q_{\sigma} \int \mathbf{v} \frac{\partial}{\partial \mathbf{v}} \cdot [(\mathbf{E} + \mathbf{v} \times \mathbf{B}) f_{\sigma}] = 0; \quad (2.67)$$

the right-hand side is zero since $\mathbf{R}_{ei} + \mathbf{R}_{ie} = 0$, i.e., the total plasma cannot exert drag on itself. We now define random velocities relative to $\mathbf{U}$ (rather than to $\mathbf{u}_{\sigma}$ as was the case for the two-fluid equations) so that the second term can be written as

$$\sum_{\sigma} \int m_{\sigma} \mathbf{v} \mathbf{v} f_{\sigma} d\mathbf{v} = \sum_{\sigma} \int m_{\sigma} (\mathbf{v}' + \mathbf{U})(\mathbf{v}' + \mathbf{U}) f_{\sigma} d\mathbf{v} = \sum_{\sigma} \int m_{\sigma} \mathbf{v}' \mathbf{v}' f_{\sigma} d\mathbf{v} + \rho \mathbf{U} \mathbf{U}, \quad (2.68)$$

where $\sum_{\sigma} \int m_{\sigma} \mathbf{v}' f_{\sigma} d\mathbf{v} = 0$ has been used to eliminate terms linear in $\mathbf{v}'$. The MHD pressure tensor is now defined in terms of the random velocities relative to $\mathbf{U}$ and is given by

$$\overleftrightarrow{\mathbf{P}}^{\text{MHD}} = \sum_{\sigma} \int m_{\sigma} \mathbf{v}' \mathbf{v}' f_{\sigma} d\mathbf{v}. \quad (2.69)$$

We insert Eqs. (2.68) and (2.69) in Eq. (2.67), integrate by parts on the acceleration term, and perform the summation over species to obtain the MHD equation of motion

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) = \left(\sum_{\sigma} n_{\sigma} q_{\sigma} \right) \mathbf{E} + \mathbf{J} \times \mathbf{B} - \nabla \cdot \overleftrightarrow{\mathbf{P}}^{\text{MHD}}. \quad (2.70)$$

Magnetohydrodynamics is typically used to describe phenomena having spatial scales large enough for the plasma to be essentially neutral, i.e., $\sum_{\sigma} n_{\sigma} q_{\sigma}$ is assumed to be so small that the $(\sum_{\sigma} n_{\sigma} q_{\sigma}) \mathbf{E}$ term can be dropped from Eq. (2.70). Just as in the two-fluid situation, the left-hand side of Eq. (2.70) contains a factor times the MHD continuity equation, since the left-hand side can be expanded as

$$\begin{aligned} \frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) &= \underbrace{\left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) \right]}_{\text{zero}} \mathbf{U} + \rho \frac{\partial \mathbf{U}}{\partial t} + \rho \mathbf{U} \cdot \nabla \mathbf{U} \\ &= \rho \left(\frac{\partial \mathbf{U}}{\partial t} + \mathbf{U} \cdot \nabla \mathbf{U} \right). \end{aligned} \quad (2.71)$$

Using Eq. (2.71) in Eq. (2.70) leads to the standard form for the MHD equation of motion,

$$\rho \frac{D\mathbf{U}}{Dt} = \mathbf{J} \times \mathbf{B} - \nabla \cdot \overleftrightarrow{\mathbf{P}}^{\text{MHD}}, \quad (2.72)$$

where

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{U} \cdot \nabla \quad (2.73)$$

is the convective derivative defined using the center-of-mass velocity. Scalar approximations of the MHD pressure tensor will be postponed until after discussing the MHD Ohm's law and its implications.

2.6.3 MHD Ohm's law

Equation (2.72) provides one equation relating $\mathbf{J}$ and $\mathbf{U}$; let us now find the other one. In order to do this, consider the two-fluid electron equation of motion,

$$m_e \frac{d\mathbf{u}_e}{dt} = -e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) - \frac{1}{n_e} \nabla (n_e \kappa T_e) - v_{ei} m_e (\mathbf{u}_e - \mathbf{u}_i). \quad (2.74)$$

In MHD we are interested in low-frequency phenomena with large spatial scales. If the characteristic time scale of the phenomenon is long compared to the electron cyclotron motion, then the electron inertia term $m_e d\mathbf{u}_e/dt$ can be dropped since it is small compared to the magnetic force term $-e(\mathbf{u}_e \times \mathbf{B})$. This assumption is reasonable for velocities perpendicular to $\mathbf{B}$, but can be a poor approximation for the velocity component parallel to $\mathbf{B}$, since parallel velocities do not provide a magnetic force. Since $\mathbf{u}_e - \mathbf{u}_i = -\mathbf{J}/n_e e$ and $\mathbf{u}_i \simeq \mathbf{U}$, Eq. (2.74) reduces to the generalized Ohm's law

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} - \frac{1}{n_e e} \mathbf{J} \times \mathbf{B} + \frac{1}{n_e e} \nabla(n_e \kappa T_e) = \eta \mathbf{J}. \quad (2.75)$$

The term $-\mathbf{J} \times \mathbf{B}/n_e e$ on the left-hand side of Eq. (2.75) is called the *Hall term* and can be neglected in either of the following two cases:

1. The pressure term in the MHD equation of motion, Eq. (2.72) is negligible compared to the other two terms, which therefore must balance, giving

$$|\mathbf{J}| \sim \omega \rho |\mathbf{U}|/|\mathbf{B}|;$$

here $\omega \sim D/Dt$ is the characteristic frequency of the phenomenon. In this case comparison of the Hall term with the $\mathbf{U} \times \mathbf{B}$ term shows that the Hall term is small by a factor $\sim \omega/\omega_{ci}$, where $\omega_{ci} = q_i B/m_i$ is the ion cyclotron frequency. Thus dropping the Hall term is justified for phenomena having characteristic frequencies small compared to ω_{ci} .

2. The electron-ion collision frequency is large compared to the electron cyclotron frequency $\omega_{ce} = q_e B/m_e$, in which case the Hall term may be dropped since it is small by a factor ω_{ce}/v_{ei} compared to the right-hand side resistive term $\eta \mathbf{J} = (m_e \nu_{ei}/n_e e^2) \mathbf{J}$.

From now on, when using MHD it will be assumed that one of these conditions is true and Hall terms will be dropped (if Hall terms are retained, the system is called Hall MHD). Typically, Eq. (2.75) will not be used directly; instead its curl will be used to provide the *induction equation*

$$-\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{U} \times \mathbf{B}) - \frac{1}{n_e e} \nabla n_e \times \nabla \kappa T_e = \nabla \times \left(\frac{\eta}{\mu_0} \nabla \times \mathbf{B} \right). \quad (2.76)$$

Usually the density gradient is parallel to the temperature gradient so that the thermal electromotive force term $(n_e e)^{-1} \nabla n_e \times \nabla \kappa T_e$ can be dropped, in which case the induction equation reduces to

$$-\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{U} \times \mathbf{B}) = \nabla \times \left(\frac{\eta}{\mu_0} \nabla \times \mathbf{B} \right). \quad (2.77)$$

The thermal term is often simply ignored in the MHD Ohm's law, which is then written as

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}; \quad (2.78)$$

this is only acceptable providing we intend to take the curl and providing $\nabla n_e \times \nabla \kappa T_e \simeq 0$.

2.6.4 Ideal MHD and frozen-in flux

If the resistive term $\eta \mathbf{J}$ is so small as to be negligible compared to the other terms in Eq. (2.78), then the plasma is said to be *ideal* or *perfectly conducting*. From the Lorentz transformation of electromagnetic theory we realize that $\mathbf{E} + \mathbf{U} \times \mathbf{B} = \mathbf{E}'$, where $\mathbf{E}'$ is the electric field observed in the frame moving with velocity $\mathbf{U}$. This implies that the magnetic flux in ideal plasmas is time-invariant in the frame moving with velocity $\mathbf{U}$, because otherwise Faraday's law would imply the existence of an electric field in the moving frame. The frozen-in flux concept is the essential "bed-rock" concept underlying ideal MHD. While this concept is often an excellent approximation, it must be kept in mind that the concept becomes invalid in situations when any one of the electron inertia, electron pressure, or Hall terms becomes important and leads to different, more complex behavior.

The frozen-in flux concept is frequently expressed in a slightly different form as a frozen-in *field* concept, i.e., it is often said that magnetic field lines are frozen into the plasma so that the plasma and magnetic field lines move together as an ensemble. While this point of view can be quite intuitive and useful, it contains some ambiguity because an ideal plasma can actually move across magnetic field lines in certain situations. These situations are such that magnetic flux is preserved within the plasma even though it is moving across field lines. Discussion of this subtlety will be deferred to Section 3.5.5.

A formal proof of the frozen-in flux property will now be established by direct calculation of the rate of change of the magnetic flux through a surface $S(t)$ bounded by a material line $C(t)$, i.e., a closed contour that moves with the plasma. This magnetic flux is

$$\Phi(t) = \int_{S(t)} \mathbf{B}(\mathbf{x}, t) \cdot d\mathbf{s} \quad (2.79)$$

and the flux changes with respect to time due to either (i) the explicit time dependence of $\mathbf{B}(t)$ or (ii) changes in the surface $S(t)$ resulting from plasma motion. The rate of change of flux is thus

$$\frac{D\Phi}{Dt} = \lim_{\delta t \rightarrow 0} \left(\frac{\int_{S(t+\delta t)} \mathbf{B}(\mathbf{x}, t+\delta t) \cdot d\mathbf{s} - \int_{S(t)} \mathbf{B}(\mathbf{x}, t) \cdot d\mathbf{s}}{\delta t} \right). \quad (2.80)$$

The displacement of a segment $d\mathbf{l}$ of the bounding contour C is $\mathbf{U}\delta t$, where $\mathbf{U}$ is the velocity of this segment. The incremental change in surface area due to this displacement is $\Delta S = \mathbf{U}\delta t \times d\mathbf{l}$. The rate of change of flux can thus be expressed as

$$\begin{aligned}\frac{D\Phi}{Dt} &= \lim_{\delta t \rightarrow 0} \frac{\int_{S(t+\delta t)} \left(\mathbf{B} + \delta t \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{s} - \int_{S(t)} \mathbf{B} \cdot d\mathbf{s}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\int_{S(t)} \left(\mathbf{B} + \delta t \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{s} + \oint_C \mathbf{B} \cdot \mathbf{U} \delta t \times d\mathbf{l} - \int_{S(t)} \mathbf{B} \cdot d\mathbf{s}}{\delta t} \\ &= \int_{S(t)} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C \mathbf{B} \cdot \mathbf{U} \times d\mathbf{l} \\ &= \int_{S(t)} \left[\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{B} \times \mathbf{U}) \right] \cdot d\mathbf{s}.\end{aligned}\quad (2.81)$$

Thus, if

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{U} \times \mathbf{B}) \quad (2.82)$$

then

$$\frac{D\Phi}{Dt} = 0 \quad (2.83)$$

so that the magnetic flux linked by any closed material line is constant. Therefore, magnetic flux is frozen into an ideal plasma because Eq. (2.77) reduces to Eq. (2.82) if $\eta = 0$. Equation (2.82) is called the ideal MHD induction equation.

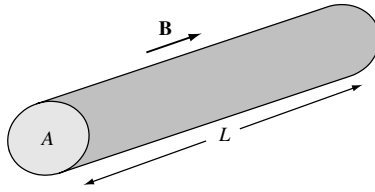
2.6.5 MHD equations of state

Double adiabatic laws

A procedure analogous to that which led to Eq. (2.35) gives the MHD adiabatic relation

$$\frac{p^{MHD}}{\rho^\gamma} = \text{const.}, \quad (2.84)$$

where again $\gamma = (N+2)/N$ and N is the number of dimensions of the system. It was shown in the previous section that magnetic flux is conserved in the plasma frame. This means that, as shown in Fig. 2.5, a tube of plasma initially occupying the same volume as a magnetic flux tube is constrained to evolve in such a way that $\int \mathbf{B} \cdot d\mathbf{s}$ stays constant over the plasma tube cross-section. For a flux tube of infinitesimal cross-section, the magnetic field is approximately uniform

Fig. 2.5 Magnetic flux tube with flux $\Phi = BA$.

over the cross-section and we may write this as $BA = \text{const.}$, where A is the cross-sectional area.

Let us define two temperatures for this magnetized plasma, namely $T_{\perp}$, the temperature corresponding to motions perpendicular to the magnetic field, and $T_{\parallel}$, the temperature corresponding to motions parallel to the magnetic field. If for some reason (e.g., anisotropic heating or compression) the temperature develops an anisotropy such that $T_{\perp} \neq T_{\parallel}$ and if collisions are infrequent, this anisotropy will persist for a long time, since collisions are the means by which the two temperatures equilibrate. Thus, rather than assuming that the MHD pressure is fully isotropic, we consider the less restrictive situation where the MHD pressure tensor is given by

$$\overleftrightarrow{\mathbf{P}}^{\text{MHD}} = \begin{bmatrix} P_{\perp} & 0 & 0 \\ 0 & P_{\perp} & 0 \\ 0 & 0 & P_{\parallel} \end{bmatrix} = P_{\perp} \overleftrightarrow{\mathbf{I}} + (P_{\parallel} - P_{\perp}) \hat{\mathbf{B}}\hat{\mathbf{B}}. \quad (2.85)$$

The first two coordinates (x, y -like) in the above matrix refer to the directions perpendicular to the local magnetic field $\mathbf{B}$ and the third coordinate (z -like) refers to the direction parallel to $\mathbf{B}$. The tensor expression on the right-hand side is equivalent (here $\overleftrightarrow{\mathbf{I}}$ is the unit tensor) but allows for arbitrary, curvilinear geometry. We now develop separate adiabatic relations for the perpendicular and parallel directions:

- Parallel direction: here the number of dimensions is $N = 1$ so that $\gamma = 3$ and so the adiabatic law gives

$$\frac{P_{\parallel}^{1D}}{\rho_{1D}^3} = \text{const.}, \quad (2.86)$$

where ρ_{1D} is the *one-dimensional* mass density; i.e., $\rho_{1D} \sim 1/L$, where L is the length along the one-dimension, e.g., along the length of the flux tube in Fig. 2.5. The three-dimensional mass density ρ , which has been used implicitly until now, has the proportionality $\rho \sim 1/LA$, where A is the cross-section of the flux tube; similarly

the three-dimensional pressure has the proportionality $P_{\parallel} \sim \rho T_{\parallel}$. However, we must be careful to realize that $P_{\parallel}^{1D} \sim \rho_{1D} T_{\parallel}$ so, using $BA = \text{const.}$, Eq. (2.86) can be recast as

$$\text{const.} = \frac{P_{\parallel}^{1D}}{\rho_{1D}^3} \sim \frac{\rho_{1D} T_{\parallel}}{\rho_{1D}^3} \sim T_{\parallel} L^2 \sim \underbrace{\left(\frac{1}{LA} \right)}_{P_{\parallel}} T_{\parallel} \underbrace{(LA)^3 B^2}_{\rho^{-3}} \quad (2.87)$$

or

$$\frac{P_{\parallel} B^2}{\rho^3} = \text{const.} \quad (2.88)$$

- Perpendicular direction: here the number of dimensions is $N = 2$ so that $\gamma = 2$ and the adiabatic law gives

$$\frac{P_{\perp}^{2D}}{\rho_{2D}^2} = \text{const.}, \quad (2.89)$$

where $P_{\perp}^{2D}$ is the 2-D perpendicular pressure, and has dimensions of energy per unit area, while ρ_{2D} is the 2-D mass density and has dimensions of mass per unit area. Thus, $\rho_{2D} \sim 1/A$ so $P_{\perp}^{2D} \sim \rho_{2D} T_{\perp} \sim T_{\perp}/A$, in which case Eq. (2.89) can be rewritten as

$$\text{const.} = \frac{P_{\perp}^{2D}}{\rho_{2D}^2} \sim T_{\perp} A \sim \left(\frac{1}{LA} \right) T_{\perp} \frac{LA}{B} \quad (2.90)$$

or

$$\frac{P_{\perp}}{\rho B} = \text{const.} \quad (2.91)$$

Equations (2.88) and (2.91) are called the double adiabatic or CGL laws after Chew, Goldberger, and Low (1956) who first developed these laws.

Single adiabatic law

If collisions are sufficiently frequent to equilibrate the perpendicular and parallel temperatures, then the pressure tensor becomes fully isotropic and the dimensionality of the system is $N = 3$ so that $\gamma = 5/3$. There is now just one pressure and one temperature and the adiabatic relation becomes

$$\frac{P}{\rho^{5/3}} = \text{const.} \quad (2.92)$$

2.6.6 MHD approximations for Maxwell's equations

The various assumptions contained in MHD lead to a simplifying approximation of Maxwell's equations. In particular, the assumption of charge neutrality in MHD makes Poisson's equation superfluous because Poisson's equation prescribes the relationship between non-neutrality and the electrostatic component of the electric

field. The assumption of charge neutrality also has implications for the current density. To see this, the two-fluid continuity equation is multiplied by q_σ and then summed over species to obtain the charge conservation equation

$$\frac{\partial}{\partial t} (\sum n_\sigma q_\sigma) + \nabla \cdot \mathbf{J} = 0. \quad (2.93)$$

Thus, charge neutrality implies

$$\nabla \cdot \mathbf{J} = 0. \quad (2.94)$$

Let us now consider Ampère's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (2.95)$$

Taking the divergence gives

$$\nabla \cdot \mathbf{J} + \varepsilon_0 \frac{\partial \nabla \cdot \mathbf{E}}{\partial t} = 0, \quad (2.96)$$

which is equivalent to Eq. (2.93) if Poisson's equation is invoked.

Finally, MHD is restricted to phenomena having characteristic velocities V_{ph} slow compared to the speed of light in vacuum, $c = (\varepsilon_0 \mu_0)^{-1/2}$. Again, t_{char} is assumed to represent the characteristic time scale for a given phenomenon and l_{char} is assumed to represent the corresponding characteristic length scale so that $V_{ph} \sim l_{char}/t_{char}$. Faraday's equation gives the scaling

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t} \implies E \sim B l_{char}/t_{char}. \quad (2.97)$$

On comparing the magnitude of the displacement current term in Eq. (2.95) to the left-hand side it is seen that

$$\frac{\mu_0 \varepsilon_0 \left| \frac{\partial \mathbf{E}}{\partial t} \right|}{|\nabla \times \mathbf{B}|} \sim \frac{c^{-2} E/t_{char}}{B/l_{char}} \sim \left(\frac{V_{ph}}{c} \right)^2. \quad (2.98)$$

Thus, if $V_{ph} \ll c$ the displacement current term can be dropped from Ampère's law resulting in the so-called "pre-Maxwell" form

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (2.99)$$

The divergence of Eq. (2.99) gives Eq. (2.94) so it is unnecessary to specify Eq. (2.94) separately.

2.7 Summary of MHD equations

We may now summarize the MHD equations:

1. Mass conservation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0. \quad (2.100)$$

2. Equation of state and associated equation of motion

- (a) Single adiabatic regime, collisions equilibrate perpendicular and parallel temperatures so that pressure and temperature are both isotropic,

$$\frac{P}{\rho^{5/3}} = \text{const.}, \quad (2.101)$$

and the equation of motion is

$$\rho \frac{D\mathbf{U}}{Dt} = \mathbf{J} \times \mathbf{B} - \nabla P. \quad (2.102)$$

- (b) Double adiabatic regime, the collision frequency is insufficient to equilibrate perpendicular and parallel temperatures so that

$$\frac{P_{\parallel} B^2}{\rho^3} = \text{const.}, \quad \frac{P_{\perp}}{\rho B} = \text{const.} \quad (2.103)$$

and the equation of motion is

$$\rho \frac{D\mathbf{U}}{Dt} = \mathbf{J} \times \mathbf{B} - \nabla \cdot \left[P_{\perp} \overleftrightarrow{\mathbf{I}} + (P_{\parallel} - P_{\perp}) \hat{\mathbf{B}} \hat{\mathbf{B}} \right]. \quad (2.104)$$

3. Faraday's law

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad (2.105)$$

4. Ampère's law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (2.106)$$

5. Ohm's law

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}. \quad (2.107)$$

These equations provide a self-consistent description of phenomena that satisfy all the various assumptions we have made, namely:

- (i) The plasma is charge-neutral since characteristic lengths are much longer than a Debye length;
- (ii) The characteristic velocity of the phenomenon under consideration is slow compared to the speed of light;
- (iii) The pressure and density gradients are parallel, so there is no electrothermal EMF;
- (iv) The time scale is long compared to both the electron and ion cyclotron periods.

Even though these assumptions are self-consistent, they may not accurately portray a real plasma and so MHD models, while intuitively appealing, must be used with caution.

Finally, it is worth mentioning that MHD plasmas can be categorized yet another way, namely according to the relative importance of the magnetic force $\mathbf{J} \times \mathbf{B}$ compared to the hydrodynamic force $\nabla \cdot \overleftrightarrow{\mathbf{P}}$. If the magnetic force is negligible compared to the hydrodynamic force, then there is not much point in using MHD because in this case the system of equations is simply classical hydrodynamics. Thus, the only non-trivial MHD situations are where the magnetic and hydrodynamic forces are of comparable importance or where the magnetic force is much more important than the hydrodynamic force. Using Amperè's law, Eq. (2.106), the nominal ratio of the hydrodynamic force to the magnetic force is defined as

$$\beta = \frac{P}{B^2/2\mu_0} \sim \frac{\nabla \cdot \overleftrightarrow{\mathbf{P}}}{\mathbf{J} \times \mathbf{B}} \sim \frac{P/L}{B^2/2\mu_0 L}; \quad (2.108)$$

the characteristic gradient scale length L is assumed to be comparable for both types of forces and so cancels out in the comparison. Low- β plasmas are those where $B^2/2\mu_0$ is much larger than P so the hydrodynamic force is negligible compared to the magnetic force, whereas $\beta = \mathcal{O}(1)$ plasmas are those where the magnetic and hydrodynamic forces are comparable.

2.8 Classical transport

Consider a cylindrical coordinate system $\{r, \theta, z\}$ with a uniform steady-state magnetic field $\mathbf{B} = B_z \hat{z}$ and suppose that an azimuthally symmetric, finite-pressure plasma exists in the region $r \leq a$ and is surrounded by vacuum. Since the right-hand side of Eq. (2.105) must vanish in steady state, if an equilibrium electric field exists, it must be electrostatic, i.e., be of the form $\mathbf{E} = -\nabla\phi$. Because of the assumed symmetry, this electrostatic electric field would also have to be azimuthally symmetric. The last two considerations mean that the θ component of the electric field would have to be zero since $E_\theta = -r^{-1}\partial\phi/\partial\theta$ and θ derivatives of all physical quantities must vanish because of the assumed steady-state azimuthal symmetry. Because $E_\theta = 0$, the θ component of the MHD Ohm's law, Eq. (2.107), is simply

$$-U_r B_z = \eta J_\theta. \quad (2.109)$$

This leads to the immediate and important conclusion that if the plasma is perfectly conducting (i.e., $\eta = 0$) then $U_r = 0$ and so the plasma will not be able to move across the magnetic field. In this case, one can say that the plasma is frozen to

the field lines, but it should be noted that this behavior relies on the existence of azimuthal symmetry.

We now consider the more general situation where the resistivity is finite, in which case there will be a finite radial motion of the plasma across the magnetic field, i.e., $U_r = -\eta J_\theta / B_z$. The azimuthal current J_θ can be determined from the steady-state r component of the MHD equation of motion, Eq. (2.104),

$$0 = J_\theta B_z - \frac{\partial P_\perp}{\partial r} \quad (2.110)$$

so, combining Eqs. (2.109) and (2.110) gives

$$U_r = -\frac{\eta}{B_z^2} \frac{\partial P_\perp}{\partial r}. \quad (2.111)$$

This gives an azimuthally symmetric, radially outward motion proportional to resistivity (i.e., to electron-ion collisions) and the radial pressure gradient, but inversely proportional to the square of the magnetic field strength. Hence, transport of plasma across magnetic fields is driven by pressure gradients, is allowed to happen if the plasma is not perfectly conducting, and is reduced if the magnetic field is strong. This process is called “classical transport.” The concept of a plasma being frozen to an azimuthally symmetric flux tube is thus valid if the magnetic field is sufficiently strong and the plasma is a sufficiently good conductor for classical transport to be negligible. Using $P_\perp = n(\kappa T_{e\perp} + \kappa T_{i\perp})$ and $\eta = m_e \nu_{ei} / ne^2$, the classical transport flux $\Gamma = nU_r$ can be expressed as

$$\Gamma_r = -D_{\text{classical}} \frac{\partial n}{\partial r}, \quad (2.112)$$

where

$$D_{\text{classical}} = \frac{m_e \nu_{ei} (\kappa T_{e\perp} + \kappa T_{i\perp})}{e^2 B_z^2} \quad (2.113)$$

is called the classical particle diffusion coefficient. This coefficient can equivalently be written as

$$D_{\text{classical}} = \left(1 + \frac{T_{i\perp}}{T_{e\perp}}\right) \frac{r_{Le}^2}{\tau_{ei}}, \quad (2.114)$$

where $r_{Le} = \sqrt{\kappa T_e / m_e \omega_{ce}^2}$ is the nominal electron-cyclotron radius and $\omega_{ce} = eB/m_e$ is the electron cyclotron frequency. Like any diffusion coefficient, the classical diffusion coefficient has the dimensions of (step-size)²/(time-step). It is intrinsically ambipolar because it has been derived in the context of MHD where the plasma is intrinsically neutral. An examination of the individual ion

and electron cross-field diffusions would show that the ion diffusion coefficient is much larger than the electron diffusion coefficient, but then a resulting ambipolar electric field would be established, which would increase the outward electron flux and decrease the outward ion flux so as to produce the ambipolar result given here. Since $r_{Le}^2 \sim T/B$ and $\tau_{ei} \sim T^{3/2}$ it is seen that $D_{classical} \sim 1/T^{1/2}B^2$ and so is very small for hot plasmas in strong magnetic fields.

2.9 Sheath physics and Langmuir probe theory

Let us now turn attention back to Vlasov theory and discuss an immediate practical application of this theory. The properties of collisionless Vlasov equilibria can be combined with Poisson's equation to develop a model for the potential in the steady-state transition region between a plasma and a conducting wall; this region is known as a sheath and is important in many practical situations. The sheath is non-neutral and its width is of the order of a Debye length. The exact sheath potential profile must be solved numerically because of the transcendental nature of the relevant equations, but a useful approximate solution can be obtained by a simple analytic argument, which will now be presented. Sheath physics is of particular importance for interpreting the current-voltage characteristics of Langmuir probes, small metal wires inserted into low-temperature plasmas for diagnostic purposes. Biasing a Langmuir probe at a sequence of voltages and then measuring the resulting current provides a simple way to gauge both the plasma density and the electron temperature.

The model presented here is the simplest possible model for sheaths and Langmuir probes and so is one dimensional. The geometry, sketched in Fig. 2.6, idealizes the Langmuir probe as a metal wall located at $x = 0$ and biased to a potential ϕ_{probe} ; this geometry could also be used to describe an actual biased metal wall at $x = 0$ in a two-dimensional plasma. The plasma is assumed to be collisionless and unmagnetized and to have an ambipolar potential ϕ_{plasma} that differs from the laboratory reference potential (so-called ground potential) because of a difference in the diffusion rates of electrons and ions out of the plasma. The plasma is assumed to extend into the semi-infinite left-hand half-plane, $-\infty < x < 0$. If $\phi_{probe} = \phi_{plasma}$, then neither electrons nor ions will be accelerated or decelerated on leaving the plasma and so each species will strike the probe (or wall) at a rate given by its respective thermal velocity. Since $m_e \ll m_i$, the electron thermal velocity greatly exceeds the ion thermal velocity. Thus, for $\phi_{probe} = \phi_{plasma}$ the electron flux to the probe (or wall) greatly exceeds the ion flux and so the current collected by the probe (or wall) will be negative.

Now consider what happens to this electron flow if the probe (or wall) is biased *negative* with respect to the plasma as shown in Fig. 2.6. To simplify the

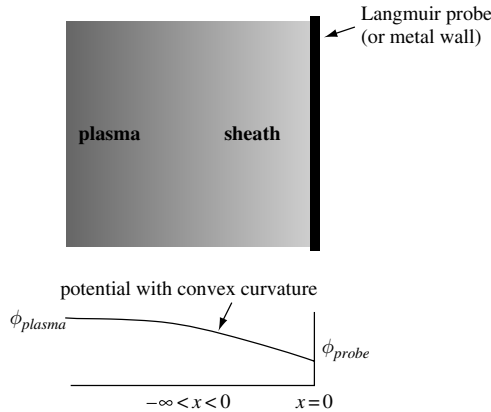


Fig. 2.6 Sketch of sheath. Top: ions are accelerated in the sheath towards the probe (wall) at $x = 0$ whereas electrons are repelled. Bottom: convex curvature of sheath requires $n_i(x)$ always to be greater than $n_e(x)$.

notation, a bar will be used to denote a potential measured relative to the plasma potential, i.e.,

$$\bar{\phi}(x) = \phi(x) - \phi_{\text{plasma}}. \quad (2.115)$$

The bias potential imposed on the probe (or wall) will be shielded out by the plasma within a distance of the order of the Debye length; this region is the sheath. The relative potential $\bar{\phi}(x)$ varies within the sheath and has the two limiting behaviors:

$$\begin{aligned} \lim_{x \rightarrow 0} \bar{\phi}(x) &= \phi_{\text{probe}} - \phi_{\text{plasma}} \\ \lim_{|x| \gg \lambda_D} \bar{\phi}(x) &= 0. \end{aligned} \quad (2.116)$$

Inside the plasma, i.e., for $|x| \gg \lambda_D$, it is assumed that the electron distribution function is Maxwellian with temperature T_e . Since the distribution function depends only on constants of the motion, the one-dimensional electron velocity distribution function must depend only on the electron energy $mv^2/2 + q_e \bar{\phi}(x)$, a constant of the motion, and so must be of the form

$$f_e(v, x) = \frac{n_0}{\sqrt{2\pi\kappa T_e/m_e}} \exp\left(-\left(\frac{mv^2/2 + q_e \bar{\phi}(x)}{\kappa T_e}\right)\right) \quad (2.117)$$

in order to be Maxwellian when $|x| \gg \lambda_D$.

The electron density is

$$n_e(x) = \int_{-\infty}^{\infty} dv f_e(v, 0) = n_0 e^{-q_e \bar{\phi}(x)/\kappa T_e}. \quad (2.118)$$

When the probe is biased negative with respect to the plasma, only those electrons with sufficient energy to overcome the negative potential barrier will be collected by the probe.

The ion dynamics is not a mirror image of the electron dynamics. This is because a repulsive potential prevents passage of particles having insufficient initial energy to climb over a potential barrier whereas an attractive potential allows passage of all particles entering a region of depressed potential. Particle density is thus reduced compared to the inlet density for *both* repulsive and attractive potentials but for different reasons. As shown in Eq. (2.118) a repulsive potential reduces the electron density exponentially (this is essentially the Boltzmann analysis developed in the theory of Debye shielding). Suppose the ions are cold and enter a region of attractive potential with velocity u_0 . Flux conservation shows that $n_0 u_0 = n_i(x) u_i(x)$ and since the ions accelerate to higher velocity when falling down the attractive potential, the ion density must decrease. Thus the electron density scales as $\exp(-|q_e \bar{\phi}|/\kappa T_e)$ and so decreases upon approaching the wall in response to what is a repulsive potential for electrons whereas the ion density scales as $1/u_i(x)$ and *also* decreases upon approaching the wall in response to what is an attractive potential for ions.

It is now important to recall that the probe was assumed to be biased negatively with respect to the plasma. Since quasi-neutrality within the plasma mandates that the electric field must vanish inside the plasma, the potential must have a downward slope on going from the plasma to the probe and the derivative of this slope must also be downward. This means that the potential $\bar{\phi}$ must have a convex curvature, i.e., a negative second derivative as indicated in Fig. 2.6 (bottom). However, the one-dimensional Poisson's equation

$$\frac{d^2 \bar{\phi}}{dx^2} = -\frac{e}{\epsilon_0} (n_i(x) - n_e(x)) \quad (2.119)$$

shows that in order for $\bar{\phi}$ to have a negative second derivative, it is necessary to have $n_i(x) > n_e(x)$ everywhere. This condition will now be used to estimate the inflow velocity of the ions at the location where they enter the sheath from the bulk plasma.

Ion energy conservation gives

$$\frac{1}{2} m_i u^2(x) + e \bar{\phi}(x) = \frac{1}{2} m_i u_0^2, \quad (2.120)$$

which can be solved to give

$$u(x) = \sqrt{u_0^2 - 2e\bar{\phi}(x)/m_i}. \quad (2.121)$$

Using the ion flux conservation relation $n_0 u_0 = n_i(x) u_i(x)$ the local ion density is found to be

$$n_i(x) = \frac{n_0}{(1 - 2e\bar{\phi}(x)/mu_0^2)^{1/2}}. \quad (2.122)$$

The convexity requirement $n_i(x) > n_e(x)$ implies

$$(1 - 2e\bar{\phi}(x)/m_i u_0^2)^{-1/2} > e^{e\bar{\phi}(x)/\kappa T_e}. \quad (2.123)$$

Bearing in mind that $\bar{\phi}(x)$ is negative, this can be rearranged as

$$1 + \frac{2e|\bar{\phi}(x)|}{m_i u_0^2} < e^{2e|\bar{\phi}(x)|/\kappa T_e} \quad (2.124)$$

or

$$\frac{2e|\bar{\phi}(x)|}{m_i u_0^2} < \frac{2e|\bar{\phi}(x)|}{\kappa T_e} + \frac{1}{2} \left(\frac{2e|\bar{\phi}(x)|}{\kappa T_e} \right)^2 + \frac{1}{3!} \left(\frac{2e|\bar{\phi}(x)|}{\kappa T_e} \right)^3 + \dots \quad (2.125)$$

which can only be satisfied for arbitrary $|\bar{\phi}(x)|$ if

$$u_0^2 > \kappa T_e / m_i. \quad (2.126)$$

Thus, in order to be consistent with the assumption that the probe is more negative than the plasma to keep $d^2\bar{\phi}/dx^2$ negative and hence $\bar{\phi}$ convex, it is necessary to have the ions enter the region of non-neutrality with a velocity slightly larger than the so-called “ion acoustic” velocity $c_s = \sqrt{\kappa T_e / m_i}$.

The ion current collected by the probe is given by the ion flux times the probe area, i.e.,

$$\begin{aligned} I_i &= n_0 u_0 q_i A \\ &= n_0 c_s e A. \end{aligned} \quad (2.127)$$

The electron current density incident on the probe is

$$\begin{aligned} J_e(x) &= \int_0^\infty dv q_e v f_e(v, 0) \\ &= \frac{n_0 q_e e^{-q_e \bar{\phi}(x)/\kappa T_e}}{\sqrt{\pi 2\kappa T_e / m_e}} \int_0^\infty dv v e^{-mv^2/2\kappa T_e} \\ &= n_0 q_e \sqrt{\frac{\kappa T_e}{2\pi m_e}} e^{-e|\bar{\phi}(x)|/\kappa T_e} \end{aligned} \quad (2.128)$$

and so the electron current collected at the probe is

$$I_e = -n_0 e A \sqrt{\frac{\kappa T_e}{2\pi m_e}} e^{-e|\bar{\phi}(x)|/\kappa T_e}. \quad (2.129)$$

Thus, the combined electron and ion current collected by the probe is

$$\begin{aligned} I &= I_i + I_e \\ &= n_0 c_s e A - n_0 e A \sqrt{\frac{\kappa T_e}{2\pi m_e}} e^{-e|\bar{\phi}(x)|/\kappa T_e}. \end{aligned} \quad (2.130)$$

The electron and ion currents cancel each other when

$$\sqrt{\frac{2\kappa T_e}{m_i}} = \sqrt{\frac{\kappa T_e}{2\pi m_e}} e^{-e|\bar{\phi}(x)|/\kappa T_e} \quad (2.131)$$

i.e., when

$$\begin{aligned} e|\bar{\phi}_{probe}|/\kappa T_e &= \ln \sqrt{\frac{m_i}{4\pi m_e}} \\ &= 2.5 \text{ for hydrogen.} \end{aligned} \quad (2.132)$$

This can be expressed as

$$\phi_{probe} = \phi_{plasma} - \frac{\kappa T_e}{e} \ln \sqrt{\frac{m_i}{4\pi m_e}} \quad (2.133)$$

and shows that when the probe potential is more negative than the plasma potential by an amount $T_e \ln \sqrt{m_i/4\pi m_e}$, where T_e is expressed in electron volts, then no current flows to the probe. This potential is called the floating potential, because an insulated object immersed in the plasma will always charge up to the floating potential since this is the potential at which no current flows to the object.

These relationships can be used as a simple diagnostic for the plasma density and electron temperature. If a probe is biased with a large negative potential, then no electrons are collected but an ion flux is collected. The collected current is called the ion saturation current and is given by $I_{sat} = n_0 c_s e A$. The ion saturation current is then subtracted from all subsequent measurements giving the electron current $I_e = I - I_{sat} = n_0 e A (\kappa T_e/2\pi m_e)^{1/2} \exp(-e|\bar{\phi}(x)|/\kappa T_e)$. The slope of a logarithmic plot of I_e versus ϕ gives $1/\kappa T_e$ and so can be used to measure the electron temperature. Once the electron temperature is known, c_s can be calculated. The plasma density can then be calculated from the ion saturation current measurement and knowledge of the probe area. Langmuir probe measurements are simple to implement but are not very precise, typically having an uncertainty of a factor of two or more.

2.10 Assignments

1. Prove Stirling's formula. To do this first show

$$\begin{aligned}\ln N! &= \ln 1 + \ln 2 + \ln 3 + \dots + \ln N \\ &= \sum_{j=1}^N \ln j.\end{aligned}$$

Now assume N is large and, using a graphical argument, show that the above expression can be expressed as an integral

$$\ln N! \approx \int_?^? h(x) dx,$$

where the form of $h(x)$ and the limits of integration are to be provided. Evaluate the integral and obtain Stirling's formula

$$\ln N! \simeq N \ln N - N \text{ for large } N.$$

This is an effective way of calculating the values of factorials of large numbers. Check the accuracy of Stirling's formula by evaluating the left- and right-hand sides of Stirling's formula numerically and plot the results for $N = 1, 10, 100, 1000, 10^4$ and higher if possible.

2. Variational calculus and Lagrange multipliers. Many physical problems involve finding the function $f(x)$ that maximizes or minimizes integrals of the form

$$I = \int_{x_1}^{x_2} w(f(x), x) dx \text{ or} \quad (2.134a)$$

$$I = \int_{x_1}^{x_2} w(f(x), df/dx, x) dx \quad (2.134b)$$

and there may or may not be restrictions on the value of $f(x_1)$ and $f(x_2)$. Entropy is of the form given in Eq. (2.134a), since we may identify $w = -f \ln f$, $x_1 = -\infty$, and $x_2 = +\infty$ and replace the dummy variable x by the dummy variable v so

$$S = - \int_{-\infty}^{\infty} f(v) \ln f(v) dv.$$

The total kinetic energy of a distribution of particles is also of the form given by Eq. (2.134a), since we may again use v as the dummy variable and let $w = mv^2 f(v)/2$. The Lagrange method for mechanics (to be discussed in Chapter 3) involves integrals of the form given in Eq. (2.134b). For now, let us consider the specific problem of determining the function $f(x)$ that minimizes I as given in Eq. (2.134a). We assume that such an f exists and call this minimizing function $f_{\min}$. We then consider the neighboring function

$$f(x) = f_{\min}(x) + \varepsilon \eta(x), \quad (2.135)$$

where ε is much smaller than unity and $\eta(x)$ is an *arbitrary* function of order unity. We may then write Eq. (2.134a) as

$$I = \int_{x_1}^{x_2} w(f_{\min}(x) + \varepsilon\eta(x), x) dx. \quad (2.136)$$

Because ε is small, Taylor expansion of I to second order in ε gives

$$I = \int_{x_1}^{x_2} \left\{ w(f_{\min}(x), x) + \varepsilon\eta(x) \left(\frac{\partial w}{\partial f} \right)_{f=f_{\min}} + \frac{1}{2} (\varepsilon\eta(x))^2 \left(\frac{\partial^2 w}{\partial f^2} \right)_{f=f_{\min}} \right\} dx. \quad (2.137)$$

(a) Define

$$I_{\min} = \int_{x_1}^{x_2} w(f_{\min}(x), x) dx \quad (2.138)$$

and consider $I - I_{\min}$, where I is given by Eq. (2.136). If $f_{\min}$ is indeed the function that minimizes I , explain that $I - I_{\min}$ must be positive if ε is not zero.

(b) Let the term that is first order in ε in Eq. (2.137) be called δI and the term that is second order in ε be called $\delta^2 I$ so

$$\delta I = \varepsilon \int_{x_1}^{x_2} \eta(x) \left(\frac{\partial w}{\partial f} \right)_{f=f_{\min}} dx \quad (2.139a)$$

$$\delta^2 I = \frac{\varepsilon^2}{2} \int_{x_1}^{x_2} (\eta(x))^2 \left(\frac{\partial^2 w}{\partial f^2} \right)_{f=f_{\min}} dx. \quad (2.139b)$$

Show that δI dominates $\delta^2 I$ if ε is made sufficiently small and then show that the sign of ε could then be chosen to make $I - I_{\min}$ negative unless we insist that $\delta I = 0$. Since ε is finite and $\eta(x)$ is arbitrary, argue that insisting $\delta I = 0$ requires having

$$\left(\frac{\partial w}{\partial f} \right)_{f=f_{\min}} = 0, \quad (2.140)$$

in which case

$$I - I_{\min} = \delta^2 I. \quad (2.141)$$

Since $I - I_{\min}$ must be positive and $\eta(x)$ is arbitrary what sign must $\partial^2 w / \partial f^2$ have? How is the function that minimizes I determined (hint: consider Eq. (2.140))?

(c) Now suppose that some sort of integral constraint exists, so the problem becomes one of finding the function f that minimizes I subject to the constraint

$$K = \int_{x_1}^{x_2} k(f(x), x) dx = \text{const.} \quad (2.142)$$

By inserting $f = f_{\min} + \varepsilon\eta(x)$ into the constraint equation, show that

$$\delta K = \varepsilon \int_{x_1}^{x_2} \eta(x) \frac{\partial k}{\partial f} dx = 0, \quad (2.143)$$

where $\partial k / \partial f$ is understood to be evaluated at $f = f_{\min}$.

(d) Since $\eta(x)$ is arbitrary, choose it to be

$$\eta(x) = \varepsilon \delta(x-a) + \mu \varepsilon \delta(x-b), \quad (2.144)$$

where $x_1 < a < x_2$, $x_1 < b < x_2$, $a \neq b$, μ is an arbitrary constant of order unity, and the delta function used here is defined specifically as

$$\delta(x) = \begin{cases} \frac{1}{\varepsilon} & \text{if } |x| < \varepsilon/2 \\ 0 & \text{if } |x| \geq \varepsilon/2. \end{cases} \quad (2.145)$$

Show that these definitions are consistent with the requirement that $\eta(x)$ is of order unity and arbitrary.

(e) By using Eq. (2.144) in Eqs. (2.139a) and (2.143) obtain the coupled equations

$$\begin{aligned} \left(\frac{\partial w}{\partial f} \right)_{x=a} + \mu \left(\frac{\partial w}{\partial f} \right)_{x=b} &= 0 \\ \left(\frac{\partial k}{\partial f} \right)_{x=a} + \mu \left(\frac{\partial k}{\partial f} \right)_{x=b} &= 0, \end{aligned} \quad (2.146)$$

where f , if it appears explicitly, is understood to be $f_{\min}$, i.e., the f that minimizes I subject to the constraint $K = \text{const.}$

(f) Define the constant λ to be

$$\lambda = \frac{\left(\frac{\partial w}{\partial f} \right)_{x=a}}{\left(\frac{\partial k}{\partial f} \right)_{x=a}} \quad (2.147)$$

and use Eqs. (2.146) to prove that

$$\frac{\left(\frac{\partial w}{\partial f} \right)_{x=b}}{\left(\frac{\partial k}{\partial f} \right)_{x=b}} = \lambda. \quad (2.148)$$

(g) Using the property that b is arbitrary show that

$$\frac{\partial w}{\partial f} - \lambda \frac{\partial k}{\partial f} = 0 \quad (2.149)$$

must be true for any x .

(h) Use the preceding results to explain why finding the f that minimizes I subject to the constraint $K = \text{const.}$ is equivalent to finding the f that minimizes $I - \lambda K$ where λ is a constant. The constant λ is called a Lagrange multiplier.

(i) Show that if there are two constraints, say $K = \text{const.}$ and $L = \text{const.}$, then minimization of I subject to these two constraints is equivalent to minimizing $I - \lambda_1 K - \lambda_2 L$, where λ_1 and λ_2 are two independent constants. (Hint: do this in two stages).

(j) Show that maximizing S is equivalent to minimizing $-S$.

3. Suppose that a group of N particles with charge q_σ and mass m_σ are located in an electrostatic potential $\phi(\mathbf{x})$. What is the maximum entropy distribution function for this situation (give a derivation)?
4. Prove that

$$\int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}. \quad (2.150)$$

Hint: consider the integral

$$\int_{-\infty}^{\infty} dx e^{-x^2} \int_{-\infty}^{\infty} dy e^{-y^2} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy e^{-(x^2+y^2)},$$

and note that $dx dy$ is an element of area. Then, instead of using Cartesian coordinates for the integral over area, use cylindrical coordinates and express all quantities in the double integral in cylindrical coordinates.

5. Evaluate the integrals

$$\int_{-\infty}^{\infty} dx x^2 e^{-ax^2}, \int_{-\infty}^{\infty} dx x^4 e^{-ax^2}.$$

Hint: take the derivative of both sides of Eq. (2.150) with respect to a .

6. Suppose that a particle starts at time $t = t_0$ with velocity $\mathbf{v}_0$ at location $\mathbf{x}_0$ and is located in a uniform, constant magnetic field $\mathbf{B} = B \hat{z}$. There is no electric field. Calculate its position and velocity as a function of time. Make sure your solution satisfies the initial conditions on both velocity and position. Be careful to treat motion parallel to the magnetic field as well as perpendicular. Express your answer in vector form as much as possible; use the subscripts $\parallel, \perp$ to denote directions parallel and perpendicular to the magnetic field, and use $\omega_c = qB/m$ to denote the cyclotron frequency. Show that $f(\mathbf{x}_0)$ is a solution of the Vlasov equation.
7. Thermal force (Braginskii 1965). If there is a temperature gradient then because of the temperature dependence of collisions, there turns out to be a subtle additional drag force, which is proportional to ∇T . To find this force, suppose a temperature gradient exists in the x direction, and consider the frictional drag on electrons passing a point $x = x_0$. The electrons moving to the right (positive velocity) at x_0 have traveled without collision from the point $x_0 - l_{mfp}$, where the temperature was $T(x_0 - l_{mfp})$, while those moving to the left (negative velocity) will have come collisionlessly from the point $x_0 + l_{mfp}$ where the temperature is $T(x_0 + l_{mfp})$. Suppose that both electrons and ions have no mean velocity at x_0 ; i.e., $u_e = u_i = 0$. Show that the total drag force on all the electrons at x_0 is

$$R_{thermal} = -2m_e n_e l_{mfp} \frac{\partial}{\partial x} (v_{ei} v_{Te}).$$

Normalize the collision frequency, thermal velocity, and mean free paths to their values at $x = x_0$, where $T = T_0$; e.g., $v_{Te}(T) = v_{Te0}(T/T_0)^{1/2}$. By writing $\partial/\partial x = (\partial T/\partial x) \partial/\partial T$ and using these normalized values show that

$$\mathbf{R}_{thermal} = -2n_e \kappa \nabla T_e.$$

A more accurate treatment that does a proper averaging over velocities gives

$$\mathbf{R}_{\text{thermal}} = -0.71 n_e \kappa \nabla T_e.$$

8. MHD with neutrals. Suppose a plasma is partially (perhaps weakly) ionized so that besides moment equations for ions and electrons there will also be moment equations for neutrals. Now the constraints will be different, since ionization and recombination will genuinely create and annihilate both plasma particles and neutrals. Construct a set of constraint equations on the collision operators, which now include ionization and recombination as well as scattering. Take the zeroth and first moment of the three Vlasov equations for ions, electrons, and neutrals and show that the continuity equation is formally the same as before, i.e.,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0,$$

providing ρ refers to the total mass density of the entire fluid (electrons, ions, and neutrals) and $\mathbf{U}$ refers to the center-of-mass velocity of the entire fluid. Show also that the equation of motion is formally the same as before, provided the pressure refers to the pressure of the entire configuration:

$$\rho \frac{D\mathbf{U}}{Dt} = -\nabla P + \mathbf{J} \times \mathbf{B}.$$

Show that Ohm's law will be the same as before, providing the plasma is sufficiently collisional so that the Hall term can be dropped, and so Ohm's law is

$$\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}$$

Explain how the neutral component of the plasma gets accelerated by the $\mathbf{J} \times \mathbf{B}$ force – this must happen since the inertial part of the equation of motion (i.e., $\rho D\mathbf{U}/Dt$) includes the acceleration on neutrals. Assume electron temperature gradients are parallel to electron density gradients so that the electrothermal force can be ignored.

9. MHD Heat Transport Equation. Define the MHD N -dimensional pressure

$$P = \frac{1}{N} \sum_{\sigma} m_{\sigma} \int \mathbf{v}' \cdot \mathbf{v}' f_{\sigma} d^N \mathbf{v},$$

where $\mathbf{v}' = \mathbf{v} - \mathbf{U}$ and N is the number of dimensions of motion (e.g., if motion in only one dimension is considered, then $N = 1$, and $\mathbf{v}$, $\mathbf{U}$ are one dimensional, etc.). Also define the isotropic MHD heat flux

$$\mathbf{q} = \sum_{\sigma} m_{\sigma} \int \mathbf{v}' \frac{v'^2}{2} f d^N \mathbf{v}.$$

- (i) By taking the second moment of the Vlasov equation for each species (i.e., use $v^2/2$) and summing over species show

$$\frac{N}{2} \frac{DP}{Dt} + \frac{N+2}{2} P \nabla \cdot \mathbf{U} = -\nabla \cdot \mathbf{q} + \mathbf{J} \cdot (\mathbf{E} + \mathbf{U} \times \mathbf{B}).$$

Table 2.1

	a	B	n	T_e, T_i	$t \sim \frac{a^2}{D}$
	(m)	(T)	(m ⁻³)	(eV)	(s)
solar corona loop	10 ⁶	10 ⁻²	10 ¹⁵	100	?
fusion-level tokamak	2	5	10 ²⁰	10 ⁴	?
nominal lab discharge	0.05	0.1	10 ¹⁸	5 – 10	?

Hints:

- (a) Prove that $\mathbf{U} \cdot (m_\sigma \sum \int \mathbf{v}' \mathbf{v}' f_\sigma d^N \mathbf{v}) = P \mathbf{U}$ assuming that f_σ is isotropic.
- (b) What happens to $\sum_\sigma m_\sigma \int v^2 C_{\sigma\alpha} f_\sigma d^N \mathbf{v}$?
- (c) Using the momentum and continuity equations prove that

$$\frac{\partial}{\partial t} \left(\frac{\rho U^2}{2} \right) + \nabla \cdot \left(\frac{\rho U^2}{2} \mathbf{U} \right) = -\mathbf{U} \cdot \nabla P + \mathbf{U} \cdot (\mathbf{J} \times \mathbf{B}).$$

- (ii) Using the continuity equation and Ohm’s law show that

$$\frac{N}{2} \frac{D P}{D T} - \frac{N+2}{2} \frac{P}{\rho} \frac{D \rho}{D t} = -\nabla \cdot \mathbf{q} + \eta J^2.$$

Show that if both the heat flux term $-\nabla \cdot \mathbf{q}$ and the Ohmic heating term ηJ^2 can be ignored, then the pressure and density are related by the adiabatic condition $P \sim \rho^\gamma$, where $\gamma = (N+2)/N$. By assuming $D/Dt \sim \omega$ and $\nabla \sim k$ show that the dropping of these two right-hand terms is equivalent to assuming $\omega \gg \nu_{ei}$ and $\omega/k \gg v_T$. Explain why the phenomenon should be isothermal if $\omega/k \ll v_T$.

- 10. Classical diffusion. Show by dimensional analysis that the nominal distance diffused in a time t is given by $x^2 \sim Dt$ (do this by combining Fick’s law $\Gamma = -D \nabla n$ and the continuity equation). Using the parameters given in Table 2.1 for a solar corona loop, a tokamak with fusion-level parameters, and a nominal laboratory discharge, calculate the time required for the plasma to diffuse out of the configuration if the diffusion is classical. In other words, fill in the last column of the table using Eq. (2.114) to calculate the nominal time (or range of times) to diffuse a distance a , where a is the nominal radius of the configuration. Are these times long or short compared to what is of interest?
- 11. Sketch the current collected by a Langmuir probe as a function of the bias voltage and indicate the ion saturation current, the exponentially changing electron current, the floating potential, and the plasma potential. Calculate the ion saturation current collected by a 1 cm long, 0.25 mm diameter probe immersed in a 5 eV argon plasma that has a density $n = 10^{16} \text{ m}^{-3}$. Calculate the electron saturation current also (i.e., the current when the probe is at the plasma potential). What is the offset of the floating potential relative to the plasma potential?